Slepian Transform Based Detectors for HF Skywave Massive MIMO-OFDM Systems

Linfeng Song, *Graduate Student Member, IEEE*, Ding Shi, *Graduate Student Member, IEEE*, Lu Gan, *Senior Member, IEEE*, and Xiqi Gao, *Fellow, IEEE*

***Abstract*—In this article, we investigate signal detection for high frequency (HF) skywave massive multiple-input multiple-output (MIMO) systems with orthogonal frequency division multiplexing (OFDM) modulation. We first introduce the beam based channel model for HF skywave massive MIMO-OFDM channels and show the relationship of sparse supports between the beam domain channel and the Fourier spectrum of the space domain channel. Based on the sparse beam domain channel, we propose a separate Slepian transform (SST) based detector, where a set of modulated Slepian sequences are designed independently for user terminals (UTs). Before the minimum mean-squared error (MMSE) detection, the Slepian transform is performed for each UT to reduce the dimension of the observation vector and the channel matrix, thus avoiding high dimensional matrices multiplications and inversions. Due to the sparsity of the beam domain channel, the Slepian transform of channel vectors can be efficiently implemented by low-dimensional matrix-vector multiplications. To further reduce the computational complexity, we propose a joint Slepian transform (JST) based detector, where a fixed set of modulated Slepian sequences are designed. As a result, Slepian transforms of the observation vector can be efficiently implemented using low-dimensional fast Fourier transform (FFT). Simulation results demonstrate the advantages of high performance and low complexity of proposed detectors.**

***Index Terms*—Massive MIMO-OFDM, HF skywave communications, Slepian transform, detector.**

I. Introduction

THE high frequency (HF) band communication, whose frequency range is typically from 3 to 30 MHz, can be utilized for over-the-horizon communications by ionosphere refraction, efficiently expanding wireless network coverage and providing global services encompassing remote places. In comparison to satellite communication, which is a research hotspot for global coverage, HF skywave communication has a comparatively low deployment cost. However, it suffers from low data rates because of the limited spectrum resource and challenging ionosphere conditions [1], [2]. By deploying a large number of antennas at the base station (BS) to serve enormous user terminals (UTs) simultaneously [3], [4], [5], massive multiple-input multiple-output (MIMO) has been exploited to significantly enhance the system sumrate and spectrum efficiencies of wireless cellular communications and has been introduced into HF skywave communications recently [6].

However, due to the increased number of BS antennas and UTs, uplink (UL) signal detection is a challenge for massive MIMO systems. Generally, massive MIMO detectors can be categorized as linear and nonlinear ones. Some nonlinear detection methods, such as local search [7] and sphere decoder [8], can generally achieve near-optimal performance while suffering from unaffordable computational complexity. On the other hand, linear detectors such as the zero-forcing (ZF) detector, minimum mean-squared error (MMSE) detector [9], [10] and their variant ones [11], [12], [13], [14] can be applied. However, when the dimension of the massive MIMO system is large, the implementation complexity of these linear detectors becomes expensive due to high-dimensional matrix operations.

Motivated by the sparsity of the beam domain channel, several beam based detection methods with low complexity have been proposed recently. In [15], the size of the channel matrix is reduced by beam selection, and ZF detection is performed. Moreover, a beam domain local MMSE algorithm is developed in [16] by using a small window in the beam domain to demodulate each UT. As a combination of the beam domain and the space domain, a layered BP detector with a post-stage outer beamformer (OBF) is proposed [17]. Thanks to the small angle spread in HF skywave massive MIMO scenario, the sparse beam domain channel can also be exploited to design detectors. However, different from the above works, in HF skywave massive MIMO communications, the transformation of the space domain channel and the beam domain channel cannot be performed directly by discrete Fourier transform (DFT). Nevertheless, the information of the beam domain channel can still be utilized for designing detectors with low complexity.

According to the beam based HF skywave massive MIMO channel model [6], each UT occupies a few non-zero beams, implying that the space domain channel is approximately band-limited and can be compressed into a low-dimensional

Manuscript received 23 August 2022; revised 15 February 2023; accepted 1 May 2023. Date of publication 10 May 2023; date of current version 17 October 2023. This article was presented in part at the 14th International Conference on Wireless Communications and Signal Processing [DOI: 10.1109/WCSP55476.2022.10039433]. This work was supported in part by the National Key R&D Program of China under Grant 2018YFB1801103, in part by the Jiangsu Province Basic Research Project under Grant BK20192002, and in part by the Key R&D Plan of Jiangsu Province under Grant BE2022067, and the Huawei Cooperation Project. The review of this article was coordinated by Prof. Hayssam Dahrouj. *(Linfeng Song and Ding Shi contributed equally to this work.) (Corresponding author: Xiqi Gao.)*

Linfeng Song, Ding Shi, and Xiqi Gao are with the National Mobile Communications Research Laboratory, Southeast University, Nanjing 210096, China, and also with Purple Mountain Laboratories, Nanjing 211111, China (e-mail: songlf@seu.edu.cn; shiding@seu.edu.cn; xqgao@seu.edu.cn).

Lu Gan is with the Department of Electronic and Computer Engineering, Brunel University London, UB8 3PH London, U.K. (e-mail: lu.gan@brunel.ac.uk).

Digital Object Identifier 10.1109/TVT.2023.3274816

0018-9545 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Southeast University. Downloaded on January 18,2024 at 06:05:40 UTC from IEEE Xplore. Restrictions apply.

form, benefiting detection complexity reduction. Inspired by the property of maximizing spectral concentration, Slepian sequences (also known as index-limited discrete prolate spheroidal sequences) [18] can be utilized for dimensionality reduction. In reality, Slepian sequences have been extensively investigated in a plethora of communication applications, e.g., receive antenna selection (AS) [19], beamforming [20], channel equalization [21], and channel estimation [22], [23], [24], [25], [26], [27].

In this article, we investigate Slepian transform based detectors for HF skywave massive MIMO systems, where the transform matrix is constructed by modulated Slepian sequences. The main contributions are summarized as follows.

- We first present a sparse beam based channel model and establish the relationship of sparse supports between the beam domain channel and the Fourier spectrum of the space domain channel, which is critical for detector designs.
- Based on the channel model, we propose a separate Slepian transform (SST) based detector where the SST is performed for each UT individually before the low-dimensional MMSE detection. Specifically, we prove that under certain conditions, the detection result of the SST based detector asymptotically equals that of the MMSE detector, and the detector can be simplified by ignoring some extremely small inter-UT interferences. Furthermore, parameters of Slepian sequences are designed accordingly. The SST based detector offers similar detection performance with much less design complexity than the MMSE detector thanks to dimensionality reduction and the use of the sparse beam domain channel.
- To further reduce the computational complexity of the SST based detector, we propose a joint Slepian transform (JST) based detector where the Slepian transform is performed jointly for UTs in a given UT set. In this case, a predetermined set of modulated Slepian sequences that are irrelevant to channel variations are designed for promising detection performance with low complexity. Particularly, the Slepian transform for the observation vector can be efficiently implemented by low-dimensional fast Fourier transform (FFT), which significantly contributes to the complexity reduction. Further, we analyze the computational complexities of proposed detectors and compared them with that of the MMSE detector. It is demonstrated that the complexity advantage of proposed detectors becomes obvious when the UTs number grows large, especially for the JST based detector.

The rest of this paper is organized as follows. In Section II, we introduce the beam based channel model for HF skywave massive MIMO-OFDM systems. In Section III, we discuss the MMSE detector and Slepian sequences. The SST based and the JST based detector are then derived using the sparsity of the beam domain channel, and the complexities of proposed detectors are discussed. Simulation results are presented in Section IV, and the work of this paper is concluded in Section V.

Notations: Boldface lower (upper) case letters denote vectors (matrices). The operators $(\cdot)^{\mathrm{H}}$, $(\cdot)^*$, $(\cdot)^{\mathrm{T}}$ denote the conjugate-transpose, conjugate, transpose of a matrix or a vector, respectively. $\mathbf{1}_M$ ($\mathbf{0}_M$), $\mathbf{I}_M$, $\mathbf{O}_{M\times N}$ and $\mathbf{e}_{M,i}$ represent M-dimensional column vector with all 1 s (0 s), M-dimensional identity matrix, $M \times N$ all-zero matrix and the i-th column of $\mathbf{I}_M$, respectively. $[\mathbf{a}]_i$, $[\mathbf{A}]_{i,j}$ and $\mathrm{vec}\{\mathbf{A}\}$ denote the i-th element of $\mathbf{a}$, the (i,j)-th element of $\mathbf{A}$, and the vectorizing of $\mathbf{A}$, respectively. $\mathrm{diag}\{\mathbf{a}\}$ indicates the diagonal matrix with $\mathbf{a}$ along its main diagonal. $\bar{\jmath} = \sqrt{-1}$ denotes the imaginary unit. $|\cdot|$ represents the absolute value for scalar and cardinality for a set. We employ $\otimes$ and $\odot$ to denote the Kronecker product operator and the Hadamard product operator, respectively. $\mathbb{E}\{\cdot\}$ denotes statistical expectation. $\delta(\cdot)$, $\langle\cdot\rangle_N$ and $\|\cdot\|$ indicate the Dirac delta function, modulo-N operation and the Frobenius norm, respectively. $\lfloor x \rfloor$ denotes the largest integer that is not greater than x. $\mathcal{CN}(\mathbf{a},\mathbf{A})$ represents the circular symmetric complex Gaussian distribution with mean $\mathbf{a}$ and covariance $\mathbf{A}$.

II. System Model

In this section, we first present the system configuration of HF skywave massive MIMO-OFDM systems. Then, we introduce the beam based channel model using sampled steering vectors, and reveal the relationship of sparse supports between the beam domain channel and the Fourier spectrum of the space domain channel.

A. System Configuration

We consider a HF skywave massive MIMO-OFDM system operating in time-division duplex (TDD) mode. The BS is equipped with a uniform linear array (ULA) with M antennas, and U single-antenna UTs are served. Actually, ULA is a reasonable configuration in the HF band since multiple antennas in the elevation direction are challenging to implement due to the decametric wavelength, resulting in typically large antenna array aperture of the BS [28], [29]. Nevertheless, the antenna array size is far smaller compared with the propagation distance. Let N_c, N_g, N_v and Δf denote the number of subcarriers, the length of the cyclic prefix (CP), the number of valid subcarriers and subcarrier spacing, respectively. Thus, the OFDM sample interval is $T_{\mathrm{sym}} = \frac{N_c+N_g}{N_c\Delta f}$. The antennas index set and UTs index set are represented as $\mathcal{A} = \{1,2,\ldots,M\}$ and $\mathcal{U} = \{1,2,\ldots,U\}$, respectively.

In HF communication systems, due to the complicated conditions of ionosphere, the carrier frequency f_c varies accordingly [30], [31]. Hereafter, we define the highest system operating frequency as f_o. The inter-antenna spacing is set as $d = \lambda_o/2$ instead of half wavelength of carrier in traditional massive MIMO communications, in which $\lambda_o = c/f_o$ is the wavelength and c is the speed of light.

B. Channel Model

In the uplink (UL) transmission, let $x_{u,n,k}$ denote the data transmitted by the UT u at the k-th subcarrier on the n-th OFDM symbol. The channel states are assumed to keep constant in one OFDM symbol. After OFDM demodulation, the received signal at the k-th subcarrier on the n-th OFDM symbol at the m-th antenna is given as

$$y_{n,k}[m] = \sum_{u\in\mathcal{U}} h_{u,n,k}[m]x_{u,n,k} + z_{n,k}[m], \tag{1}$$

Authorized licensed use limited to: Southeast University. Downloaded on January 18,2024 at 06:05:40 UTC from IEEE Xplore. Restrictions apply.

where $z_{n,k}[m] \sim \mathcal{CN}(0, \sigma_z^2)$ is the additive white Gaussian noise, $h_{u,n,k}[m]$ is the channel frequency response at the k-th subcarrier on the n-th OFDM symbol between the u-th UT and the m-th antenna at the BS, which can be expressed as [6]

$$h_{u,n,k}[m] = \sum_{p=1}^{P_u} \beta_{u,p} e^{\bar{\jmath}\phi_{u,p}} e^{\bar{\jmath}2\pi f_{u,p}^{\mathrm{d}} n T_{\mathrm{sym}}} e^{-\bar{\jmath}2\pi k \Delta f \tau_{u,p}} \cdot e^{-\bar{\jmath}2\pi (f_c + k\Delta f)\Delta\tau (m-1)\Omega_{u,p}}, \quad (2)$$

where P_u is the number of paths from the u-th UT to the BS and $\Delta\tau = d/c$. Additionally, $\beta_{u,p}$, $\phi_{u,p}$, $f_{u,p}^{\mathrm{d}}$, $\tau_{u,p}$ and $\Omega_{u,p}$ represent the gain, the initial phase, the Doppler frequency, the propagation delay to the first antenna at the BS and the direction cosine of the p-th path, respectively. We define the steering vector pointing towards Ω at the k-th subcarrier as

$$\mathbf{v}_k(\Omega) \triangleq \left[1, e^{-\bar{\jmath}2\pi(f_c+k\Delta f)\Delta\tau 1\Omega}, \ldots, e^{-\bar{\jmath}2\pi(f_c+k\Delta f)\Delta\tau(M-1)\Omega}\right]^{\mathrm{T}}. \quad (3)$$

Note that due to the spatial-wideband effect [32], the steering vector varies in different subcarriers, then the space domain channel vector $\mathbf{h}_{u,n,k} \triangleq [h_{u,n,k}[1], \ldots, h_{u,n,k}[M]]^{\mathrm{T}}$ can be expressed as

$$\mathbf{h}_{u,n,k} = \sum_{p=1}^{P_u} \eta_{u,p,n,k} \mathbf{v}_k(\Omega_{u,p}), \quad (4)$$

where $\eta_{u,p,n,k} \triangleq \beta_{u,p} e^{\bar{\jmath}\phi_{u,p}} e^{\bar{\jmath}2\pi f_{u,p}^{\mathrm{d}} n T_{\mathrm{sym}}} e^{-\bar{\jmath}2\pi k \Delta f \tau_{u,p}}$.

To represent the channel in the beam domain, we perform uniform sampling for direction cosine as $[-1, 1) = \bigcup_{\tilde{m}=1}^{\tilde{M}} \mathcal{S}_{\tilde{m}}$, where $\tilde{M} = \lceil FM\frac{f_c}{f_o} \rceil \geq M$ is the sampling number and F is the fine factor [33], $\mathcal{S}_{\tilde{m}} = [\Omega_{\tilde{m}}, \Omega_{\tilde{m}} + 2/\tilde{M})$ and $\Omega_{\tilde{m}} = (2\tilde{m} - 2 - \tilde{M})/\tilde{M}$. Define the set $\mathcal{P}_u \triangleq \{\Omega_{u,1}, \ldots, \Omega_{u,P_u}\}$ including direction cosines of all paths for UT u. Then, $\mathbf{h}_{u,n,k}$ can be approximated as [6]

$$\mathbf{h}_{u,n,k} = \sum_{\tilde{m}=1}^{\tilde{M}} \sum_{\Omega_{u,p} \in \mathcal{P}_u \cap \mathcal{S}_{\tilde{m}}} \eta_{u,p,n,k} \mathbf{v}_k(\Omega_{\tilde{m}}) = \mathbf{V}_k \tilde{\mathbf{h}}_{u,n,k}, \quad (5)$$

where $\mathbf{v}_k(\Omega_{\tilde{m}})$ is the sampled steering vector, and $\mathbf{V}_k \triangleq [\mathbf{v}_k(\Omega_1), \ldots, \mathbf{v}_k(\Omega_{\tilde{M}})]$. Each sampled steering vector in the channel representation of (5) corresponds to a physical beam in the space domain, and all of the UTs share the same set of sampled steering vectors. Therefore, it is termed as the beam based channel model. $\tilde{\mathbf{h}}_{u,n,k} \in \mathbb{C}^{\tilde{M}}$ with $[\tilde{\mathbf{h}}_{u,n,k}]_{\tilde{m}} = \sum_{\Omega_{u,p} \in \mathcal{P}_u \cap \mathcal{S}_{\tilde{m}}} \eta_{u,p,n,k}$ is the beam domain channel vector and $\mathbf{V}_k$ is called as the beam matrix at the k-th subcarrier. It is noteworthy that the beam matrix $\mathbf{V}_k$ is associated with subcarriers due to the spatial-wideband effect.

Interestingly, the beam based channel model (5) can be regarded as an extension of traditional massive MIMO channel model. When $f_o = f_c$, $F = 1$, and the spatial-wideband effect is ignored, the beam matrix $\mathbf{V}_k$ will be reduced to a unitary matrix, which is consistent with the widely used beamspace concept in [34] and [35].

We now focus on the Fourier spectrum of $h_{u,n,k}[m]$ to design detectors with low complexity. On account of the small angle spread from each UT to the BS [36], the beam domain channel vector is sparse, and is also associated with the Fourier spectrum of the space domain channel. To elaborate this, we rewrite (2) as

$$h_{u,n,k}[m] = \int_{-1}^{1} \sum_{p=1}^{P_u} \eta_{u,p,n,k} e^{-\bar{\jmath}2\pi g_k (m-1)\Omega} \delta(\Omega - \Omega_{u,p}) \mathrm{d}\Omega. \quad (6)$$

Note that f_o is the highest system operating frequency, then we assume $f_c + k\Delta f < f_o$, thus $g_k = (f_c + k\Delta f)\Delta\tau < 1/2$. Make a variable substitution $\nu = -g_k\Omega$, $\nu_{u,p,k} = -g_k\Omega_{u,p}$ and define

$$\tilde{h}_{u,n,k}(\nu) \triangleq \begin{cases} \sum_{p=1}^{P_u} \eta_{u,p,n,k} \delta(\nu - \nu_{u,p,k}), & \nu \in [-g_k, g_k] \\ 0, & \nu \in [-\frac{1}{2}, -g_k) \cup (g_k, \frac{1}{2}). \end{cases} \quad (7)$$

Then (6) can be rewritten as

$$\begin{aligned} h_{u,n,k}[m] &= \int_{-g_k}^{g_k} \tilde{h}_{u,n,k}(\nu) e^{\bar{\jmath}2\pi(m-1)\nu} \mathrm{d}\nu \\ &= \int_{-\frac{1}{2}}^{\frac{1}{2}} \tilde{h}_{u,n,k}(\nu) e^{\bar{\jmath}2\pi(m-1)\nu} \mathrm{d}\nu \\ &= \mathcal{F}^{-1}\left\{\tilde{h}_{u,n,k}(\nu) e^{-\bar{\jmath}2\pi\nu}\right\}, \end{aligned} \quad (8)$$

where $\mathcal{F}^{-1}\{\cdot\}$ represents the inverse of discrete-time Fourier transform (IDTFT). Due to the small angle spread,

$$\max_{p,p' \in \mathcal{P}_u} \{\Omega_{u,p} - \Omega_{u,p'}\} \ll 2, \quad (9)$$

resulting in

$$\max_{p,p' \in \mathcal{P}_u} \{\nu_{u,p,k} - \nu_{u,p',k}\} \ll 1. \quad (10)$$

Consequently, $h_{u,n,k}[m]$ can be regarded as a narrowband sequence, whose frequency support is specified as a continuous interval $\mathcal{L}_{u,k} = [\nu_{u,k}^{\mathrm{l}}, \nu_{u,k}^{\mathrm{r}}] \subset [-1/2, 1/2)$. When M tends to be enormous, $\nu_{u,k}^{\mathrm{l}}$ and $\nu_{u,k}^{\mathrm{r}}$ can be respectively approximated by

$$\nu_{u,k}^{\mathrm{l}} = -g_k \left(\frac{2}{\tilde{M}} \max\{\mathcal{B}_u\} - 1\right), \quad (11)$$

$$\nu_{u,k}^{\mathrm{r}} = -g_k \left(\frac{2}{\tilde{M}} \min\{\mathcal{B}_u\} - 1\right), \quad (12)$$

where $\mathcal{B}_u = \{\tilde{m} | [\tilde{\mathbf{h}}_{u,n,k}]_{\tilde{m}} \neq 0\} = \{\tilde{m} | [\tilde{\boldsymbol{\omega}}_u]_{\tilde{m}} \neq 0\}$ is the index set for non-zero elements of the beam domain channel vector with $B_u = |\mathcal{B}_u|$, and $\tilde{\boldsymbol{\omega}}_u \triangleq \mathbb{E}\{\tilde{\mathbf{h}}_{u,n,k} \odot \tilde{\mathbf{h}}_{u,n,k}^*\}$ is the channel coupling vector with $[\tilde{\boldsymbol{\omega}}_u]_{\tilde{m}} = \sum_{\Omega_{u,p} \in \mathcal{P}_u \cap \mathcal{S}_{\tilde{m}}} \beta_{u,p}^2$, which is independent of symbols and subcarriers. It is also viewed as the statistical channel state information (CSI).

III. Slepian Transform Based Detectors

In this section, we develop the concept for processing MMSE detection in low dimensions. We first propose the SST based

Authorized licensed use limited to: Southeast University. Downloaded on January 18,2024 at 06:05:40 UTC from IEEE Xplore. Restrictions apply.

detector where Slepian transform is performed for each UT separately before low-dimensional MMSE detection. Then, the JST based detector is designed that allows an efficient implementation with low-dimensional FFT for further complexity reduction. Finally, the computational complexities of proposed Slepian transform based detectors are analyzed and compared with that of the MMSE detector.

A. MMSE Detector

We assume that the perfect CSI is known at the BS and focus on the signal detection at the k-th subcarrier on the n-th OFDM symbol. Indexes n and k in the subscript are omitted hereafter for brevity. Note that (1) can be rewritten in vector form

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{z}, \tag{13}$$

where $\mathbf{y} \triangleq [y[1], \ldots, y[M]]^{\mathrm{T}} \in \mathbb{C}^M$, $\mathbf{H} \triangleq [\mathbf{h}_1, \ldots, \mathbf{h}_U] \in \mathbb{C}^{M\times U}$, $\mathbf{x} \triangleq [x_1, \ldots, x_U]^{\mathrm{T}} \in \mathbb{C}^U$ and $\mathbf{z} \sim \mathcal{CN}(\mathbf{0}_M, \sigma_{\mathrm{z}}^2\mathbf{I}_M)$ are the observation vector, the channel matrix, the transmit vector and the complex Gaussian noise vector, respectively. And we assume that $\mathbb{E}\{\mathbf{x}\mathbf{x}^{\mathrm{H}}\} = \sigma_{\mathrm{x}}^2\mathbf{I}_U$, where σ_{x}^2 is the average transmit power of each UT. The MMSE detector is given as

$$\begin{aligned}\mathbf{R}^{\mathrm{mmse}} &= \underset{\mathbf{R}\in\mathbb{C}^{M\times U}}{\arg\min}\, \mathbb{E}\left\{\left\|\mathbf{x} - \mathbf{R}^{\mathrm{H}}\mathbf{y}\right\|^2\right\} \\ &= \mathbf{H}\left(\mathbf{H}^{\mathrm{H}}\mathbf{H} + \frac{1}{\rho}\mathbf{I}_U\right)^{-1},\end{aligned} \tag{14}$$

where $\rho \triangleq \sigma_{\mathrm{x}}^2/\sigma_{\mathrm{z}}^2$, and the detection result is $\hat{\mathbf{x}}^{\mathrm{mmse}} = (\mathbf{R}^{\mathrm{mmse}})^{\mathrm{H}}\mathbf{y}$.

In HF skywave massive MIMO scenarios, when M and U are both enormous, $\mathbf{R}^{\mathrm{mmse}}$ and $\hat{\mathbf{x}}^{\mathrm{mmse}}$ are computationally expensive due to high-dimensional matrix inversion and multiplication. Note that the channel energy for each UT is concentrated on non-zero beams, implying that the channel matrix and the observation vector can be compressed into low-dimensional ones by exploiting the sparsity of the beam domain channel. By dimensionality reduction, the computational complexity of MMSE detection can be reduced. Most related works achieve dimensionality reduction by performing DFT directly and selecting effective beams of UTs [17], [15]. However, the spectral leakage caused by limited number of antennas at the BS is not taken into consideration in these works. It is well known that for a band-limited finite length vector, Slepian sequences offer the most compact representation in the spectral domain which effectively suppress the impact of spectral leakage. This motivates us to utilize Slepian sequences, instead of DFT basis for detector design. Before proceeding, Slepian sequences are first introduced below:

B. Slepian Sequences

Consider a discrete sequence $s[n]$ with all-zero outside the index range $0, 1, \ldots, M-1$. We expect that the energy of $s[n]$ is most concentrated in Fourier spectrum interval $[-W, +W]$ with $W < 1/2$, i.e., the following expression is maximized [37]

$$\begin{aligned}\lambda &= \frac{\int_{-W}^{+W} |\mathcal{F}\{s[n]\}|^2 \mathrm{d}\nu}{\int_{-\frac{1}{2}}^{\frac{1}{2}} |\mathcal{F}\{s[n]\}|^2 \mathrm{d}\nu} \\ &= \frac{\int_{-W}^{+W} |\sum_{n=0}^{M-1} s[n] e^{-\bar{\jmath}2\pi n\nu}|^2 \mathrm{d}\nu}{\int_{-\frac{1}{2}}^{\frac{1}{2}} |\sum_{n=0}^{M-1} s[n] e^{-\bar{\jmath}2\pi n\nu}|^2 \mathrm{d}\nu} \\ &= \frac{\mathbf{s}^{\mathrm{H}}\mathbf{B}\mathbf{s}}{\mathbf{s}^{\mathrm{H}}\mathbf{s}},\end{aligned} \tag{15}$$

where $\mathbf{s} \triangleq [s[0], \ldots, s[M-1]]^{\mathrm{T}}$, and $\mathbf{B} \in \mathbb{R}^{M\times M}$ with $[\mathbf{B}]_{i,j} \triangleq 2W\mathrm{sinc}(2W(i-j))$ is known as prolate matrix, in which $\mathrm{sinc}(2W(i-j)) \triangleq \sin(2\pi W(i-j))/2\pi W(i-j)$ is the sinc function. As $\mathbf{B}$ is Hermitian, (15) is the Rayleigh quotient, whose maximum value is the maximum eigenvalue of $\mathbf{B}$ [38]. To this end, we decompose $\mathbf{B}$ into $\mathbf{B} = \mathbf{S}\mathbf{\Lambda}\mathbf{S}^{\mathrm{H}}$, where $\mathbf{S} \triangleq [\mathbf{s}_1, \ldots, \mathbf{s}_M] \in \mathbb{R}^{M\times M}$ consists of eigenvectors $\mathbf{s}_i$ (for $i = 1, \ldots, M$) and $\mathbf{\Lambda} \triangleq \mathrm{diag}\{\lambda_1, \ldots, \lambda_M\} \in \mathbb{R}^{M\times M}$ in which λ_i (for $i = 1, \ldots, M$) are eigenvalues in descending order. Accordingly, $\mathbf{s}_1, \ldots, \mathbf{s}_M$ are defined as Slepian sequences, whose main features are given as follows [39]:

- The M eigenvectors can be normalized such that they are orthonormal and form a basis for $\mathbb{C}^M$ (or $\mathbb{R}^M$), i.e., $\mathbf{S}^{\mathrm{H}}\mathbf{S} = \mathbf{I}_M$;
- $1 > \lambda_1 > \cdots > \lambda_M > 0$. The first $\approx 2MW$ eigenvalues extremely close to 1, whereas the remaining eigenvalues tend to 0, thus the first $\approx 2MW$ Slepian sequences are sufficient to capture the sequence of length M whose energy is mostly concentrated in $[-W, +W]$.

Moreover, to describe a sequence whose energy is most concentrated in Fouerier spectrum interval $\mathcal{M}(\overline{\nu}) = [\overline{\nu} - W, \overline{\nu} + W]$ with $\overline{\nu} \neq 0$, modulated Slepian sequences are proposed. The modulated vector and the matrix composed of modulated Slepian sequences are respectively given as [40]

$$\boldsymbol{\omega}(\overline{\nu}) \triangleq \left[1, e^{\bar{\jmath}2\pi\overline{\nu}}, \ldots, e^{\bar{\jmath}2\pi\overline{\nu}(M-1)}\right]^{\mathrm{T}} \in \mathbb{C}^M, \tag{16}$$

$$\mathbf{D}(\overline{\nu}) \triangleq [\mathbf{d}_1(\overline{\nu}), \ldots, \mathbf{d}_K(\overline{\nu})] \in \mathbb{C}^{M\times K}, \tag{17}$$

where $\mathbf{d}_i(\overline{\nu}) \triangleq \boldsymbol{\omega}(\overline{\nu}) \odot \mathbf{s}_i \in \mathbb{C}^M$.

It can be checked that $\mathbf{D}(\overline{\nu})^{\mathrm{H}}\mathbf{D}(\overline{\nu}) = \mathbf{I}_K$, and $\mathbf{d}_i(\overline{\nu})$ (for $i = 1, \ldots, M$) form an orthonormal basis for $\mathbb{C}^M$. As mentioned above, the Slepian sequences can combat the effects of spectrum leakage to some extent. To demonstrate this, we generate a HF skywave massive MIMO channel sample $\mathbf{h}_u$ and project it onto Slepian basis ($\mathcal{M}(\overline{\nu}) = \mathcal{L}_u$) and normalized DFT basis, respectively. Fig. 1 plots the normalized magnitude of basis expansion coefficients. It can be observed that the spectrum leakage suppression is obvious for Slepian basis since most energy of $\mathbf{h}_u$ is concentrated on the first several coefficients, which is much superior to the DFT basis.

Moreover, owing to the spectrum leakage suppression, when $K \approx 2MW$, $\mathbf{D}(\overline{\nu})$ is sufficient to capture the sequence $a[n]$ of length M whose energy is mostly concentrated in $\mathcal{M}(\overline{\nu})$, i.e., $\mathbf{D}(\overline{\nu})\mathbf{D}(\overline{\nu})^{\mathrm{H}}\mathbf{a} \approx \mathbf{a}$, in which $\mathbf{a} \triangleq [a[0], \ldots, a[M-1]]^{\mathrm{T}}$.

Authorized licensed use limited to: Southeast University. Downloaded on January 18,2024 at 06:05:40 UTC from IEEE Xplore. Restrictions apply.

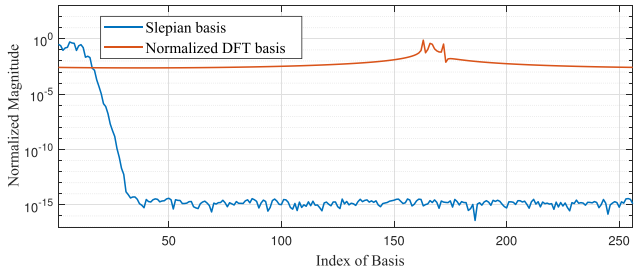


Fig. 1. Normalized magnitude of basis expansion coefficients for $\mathbf{h}_u$ with $M = 256$ and $\mathcal{L}_u = [-0.372735, -0.3294375]$.

Therefore, one can compress $\mathbf{a}$ by obtaining Slepian basis coefficients $\overline{\mathbf{a}}(\overline{\nu}) = \mathbf{D}(\overline{\nu})^{\mathrm{H}}\mathbf{a}$ instead of the M entries of $\mathbf{a}$, and $\overline{\mathbf{a}}(\overline{\nu})$ is denoted as the Slepian transform of $\mathbf{a}$, which is utilized in the subsequent detector designs.

C. Separate Slepian Transform Based Detector

In this subsection, Slepian transform is used for detection. Construct $\mathbf{D}(\overline{\nu}_u)$ with $\mathcal{M}(\overline{\nu}_u) = [\overline{\nu}_u - W, \overline{\nu}_u + W]$ for each UT u (the parameter settings of $\mathbf{D}(\overline{\nu}_u)$ are discussed later), then apply the Slepian transform to $\mathbf{y}$ as

$$\overline{\mathbf{y}}(\overline{\nu}_u) = \mathbf{D}(\overline{\nu}_u)^{\mathrm{H}}\mathbf{y} = \sum_{u' \in \mathcal{U}} \overline{\mathbf{h}}_{u'}(\overline{\nu}_u) x_{u'} + \overline{\mathbf{z}}(\overline{\nu}_u), \tag{18}$$

where $\overline{\mathbf{y}}(\overline{\nu}_u) \in \mathbb{C}^K$, $\overline{\mathbf{h}}_{u'}(\overline{\nu}_u) \triangleq \mathbf{D}(\overline{\nu}_u)^{\mathrm{H}}\mathbf{h}_{u'} \in \mathbb{C}^K$ and $\overline{\mathbf{z}}(\overline{\nu}_u) \triangleq \mathbf{D}(\overline{\nu}_u)^{\mathrm{H}}\mathbf{z} \in \mathbb{C}^K$ are the compressed observation vector, the compressed channel vector and the compressed noise vector, respectively. Note that $\mathbf{D}(\overline{\nu}_u)$ is composed of the first K columns of a unitary matrix, thus $\overline{\mathbf{z}}(\overline{\nu}_u) \sim \mathcal{CN}(0, \sigma_z^2 \mathbf{I}_K)$.

With the compressed observation vector and the compressed channel vector in (18), the MMSE detector of UT u is given as

$$\begin{aligned}\mathbf{r}_u^{\mathrm{sst}} &= \arg\min_{\mathbf{r}_u \in \mathbb{C}^K} \mathbb{E}\left\{\left|x_u - \mathbf{r}_u^{\mathrm{H}} \overline{\mathbf{y}}(\overline{\nu}_u)\right|^2\right\} \\ &= \left(\sum_{u' \in \mathcal{U}} \overline{\mathbf{h}}_{u'}(\overline{\nu}_u)\overline{\mathbf{h}}_{u'}(\overline{\nu}_u)^{\mathrm{H}} + \frac{1}{\rho}\mathbf{I}_K\right)^{-1} \overline{\mathbf{h}}_u(\overline{\nu}_u).\end{aligned} \tag{19}$$

Consequently, the detection result is

$$\hat{\mathbf{x}}^{\mathrm{sst}} = \left[\left(\mathbf{r}_1^{\mathrm{sst}}\right)^{\mathrm{H}} \overline{\mathbf{y}}(\overline{\nu}_1), \cdots, \left(\mathbf{r}_U^{\mathrm{sst}}\right)^{\mathrm{H}} \overline{\mathbf{y}}(\overline{\nu}_U)\right]^{\mathrm{T}} = \left(\mathbf{R}^{\mathrm{sst}}\right)^{\mathrm{H}} \mathbf{y}, \tag{20}$$

where the detector is expressed as

$$\mathbf{R}^{\mathrm{sst}} = \left[\mathbf{D}(\overline{\nu}_1)\mathbf{r}_1^{\mathrm{sst}}, \ldots, \mathbf{D}(\overline{\nu}_U)\mathbf{r}_U^{\mathrm{sst}}\right] \in \mathbb{C}^{M \times U}. \tag{21}$$

Generally, the detection performance of (20) is inferior to that of the MMSE detector due to the fact

$$\begin{aligned}\mathbb{E}\left\{\|\mathbf{x} - \hat{\mathbf{x}}^{\mathrm{sst}}\|^2\right\} &= \sum_{u \in \mathcal{U}} \min_{\mathbf{r}_u \in \mathbb{C}^K} \mathbb{E}\left\{\left|x_u - \mathbf{r}_u^{\mathrm{H}} \overline{\mathbf{y}}(\overline{\nu}_u)\right|^2\right\} \\ &= \sum_{u \in \mathcal{U}} \min_{\mathbf{r}_u \in \mathrm{span}\{\mathbf{D}(\overline{\nu}_u)\}} \mathbb{E}\left\{\left|x_u - \mathbf{r}_u^{\mathrm{H}} \mathbf{y}\right|^2\right\} \\ &\geq \sum_{u \in \mathcal{U}} \min_{\mathbf{r}_u \in \mathbb{C}^M} \mathbb{E}\left\{\left|x_u - \mathbf{r}_u^{\mathrm{H}} \mathbf{y}\right|^2\right\} \\ &= \mathbb{E}\left\{\|\mathbf{x} - \hat{\mathbf{x}}^{\mathrm{mmse}}\|^2\right\},\end{aligned} \tag{22}$$

where $\mathrm{span}\{\mathbf{D}(\overline{\nu}_u)\}$ represents the column space of $\mathbf{D}(\overline{\nu}_u)$. And the computation of $\mathbf{r}_u^{\mathrm{sst}}$ remains sophisticated since all UTs are included as inter-UT interference in (19). Accordingly, to guarantee the detection performance and simplify (19), we have to design appropriate parameters for $\mathbf{D}(\overline{\nu}_u)$. To address this issue, following theorems are developed, providing the criterion for designing K, W and $\overline{\nu}_u$.

Theorem 1: Assume $\mathcal{P}_u \cap \mathcal{P}_{u'} = \emptyset$ for $u \neq u'$, when $\mathcal{L}_u \subseteq \mathcal{M}(\overline{\nu}_u)$, $K = \lfloor 2MW(1+\epsilon) \rfloor$, $0 < \epsilon < \frac{1}{2W} - 1$, we have

$$\lim_{M \to \infty} \mathbb{E}\left\{\|\hat{\mathbf{x}}^{\mathrm{mmse}} - \hat{\mathbf{x}}^{\mathrm{sst}}\|^2\right\} = 0. \tag{23}$$

Proof: See Appendix A.

From Theorem 1, we know that the detection result of (20) is asymptotically equal to that of the MMSE detector if the following conditions are met: *i*) The directional cosines of the UTs are distincted and limited; *ii*) The number of antennas M is relatively large; *iii*) $\mathcal{L}_u \subseteq \mathcal{M}(\overline{\nu}_u)$ and K is slightly greater than $2MW$.

Theorem 2: When $K = \lfloor 2MW(1-\epsilon) \rfloor$, $0 < \epsilon < 1$ and $\mathcal{M}(\overline{\nu}_u) \cap \mathcal{L}_{u'} = \emptyset$, we have

$$\lim_{M \to \infty} \mathbf{D}(\overline{\nu}_u)^{\mathrm{H}} \mathbf{h}_{u'} = \mathbf{0}_K. \tag{24}$$

Proof: See Appendix B.

According to Theorem 2, when M is relatively large and K is slightly less than $\lfloor 2MW \rfloor$, if $\mathcal{M}(\overline{\nu}_u)$ and $\mathcal{L}_{u'}$ do not overlap, the columns of $\mathbf{D}(\overline{\nu}_u)$ are asymptotically orthogonal to $\mathbf{h}_{u'}$, i.e., $\overline{\mathbf{h}}_{u'}(\overline{\nu}_u) \approx \mathbf{0}_K$. Under these conditions, $\overline{\mathbf{h}}_{u'}(\overline{\nu}_u)$ can be omitted in (19) for simplification. In other words, (19) can be simplified by ignoring the negligibly small interference from UTs that are geographically remote to UT u.

With the aid of Theorem 1 and Theorem 2, the parameter settings are given as follows.

1) W and $\overline{\nu}_u$: According to Theorem 1, we set

$$W \geq \overline{W}, \quad \overline{\nu}_u = \frac{1}{2}\left(\nu_u^{\mathrm{l}} + \nu_u^{\mathrm{r}}\right), \tag{25}$$

where $\overline{W} \triangleq \frac{1}{2}\max_{u \in \mathcal{U}}\{\nu_u^{\mathrm{r}} - \nu_u^{\mathrm{l}}\}$ is half of the maximum length of $\mathcal{L}_1, \ldots, \mathcal{L}_U$. It can be checked that $\mathcal{L}_u \subseteq \mathcal{M}(\overline{\nu}_u)$ holds for each UT u. In practice, due to the effect of spectral leakage caused by limited number of antennas, we can set W a bit larger to contain the most channel energy in compressed channel vectors under acceptable detection complexity.

2) K: Under the condition $\mathcal{L}_u \subseteq \mathcal{M}(\overline{\nu}_u)$, we discuss the value selection of K. When $K \geq \lfloor 2MW \rfloor$, the detection performance is guaranteed according to Theorem 1, whilst the condition $\overline{\mathbf{h}}_{u'}(\overline{\nu}_u) \approx \mathbf{0}_K$ for $\mathcal{M}(\overline{\nu}_u) \cap \mathcal{L}_{u'} = \emptyset$ is not well satisfied according to Theorem 2, which indicates that $\overline{\mathbf{h}}_{u'}(\overline{\nu}_u)$ cannot be neglected in (19) thus leading to high complexity. Conversely, when $K < \lfloor 2MW \rfloor$, from Theorem 2, (19) can be simplified whereas the detection performance cannot be well guaranteed. Nevertheless, it is known from the simulation that the detection performance loss is vanishingly small when K is slightly below $\lfloor 2MW \rfloor$. For this reason, we compromise a small fraction of detection accuracy for complexity reduction by setting K slightly less than $2MW$. Specifically, we select K

Authorized licensed use limited to: Southeast University. Downloaded on January 18,2024 at 06:05:40 UTC from IEEE Xplore. Restrictions apply.

as the maximum integer that satisfies $\lambda_K > \lambda^{\text{th}}$, where λ^{th} is a threshold which is less than and close to 1, i.e.,

$$K = \max_{K' \in \mathcal{A}, \lambda_{K'} > \lambda^{\text{th}}} K'. \tag{26}$$

After setting parameters by (25) and (26), (19) can be simplified as

$$\mathbf{r}_u^{\text{sst}} \approx \left(\sum_{u' \in \overline{\mathcal{U}}_u} \overline{\mathbf{h}}_{u'}(\overline{\nu}_u) \overline{\mathbf{h}}_{u'}(\overline{\nu}_u)^{\text{H}} + \frac{1}{\rho} \mathbf{I}_K \right)^{-1} \overline{\mathbf{h}}_u(\overline{\nu}_u), \tag{27}$$

where

$$\overline{\mathcal{U}}_u \triangleq \left\{ u' \middle| \mathcal{M}(\overline{\nu}_u) \cap \mathcal{L}_{u'} \neq \emptyset,\ u' \in \mathcal{U} \right\} \text{ with } \overline{U}_u = |\overline{\mathcal{U}}_u|. \tag{28}$$

Additionally, since the perfect beam domain channel is obtained, $\overline{\mathbf{h}}_{u'}(\overline{\nu}_u)$ can be expressed as

$$\overline{\mathbf{h}}_{u'}(\overline{\nu}_u) = \mathbf{D}(\overline{\nu}_u)^{\text{H}} \mathbf{V} \tilde{\mathbf{h}}_{u'} = \mathbf{G}(\overline{\nu}_u) \tilde{\mathbf{h}}_{u'} = \mathbf{G}_{u'}(\overline{\nu}_u) \hat{\mathbf{h}}_{u'}, \tag{29}$$

where $\hat{\mathbf{h}}_u \triangleq \mathbf{N}_u \tilde{\mathbf{h}}_u \in \mathbb{C}^{B_u}$ represents the reduced-dimensional beam domain channel with non-zero elements of $\tilde{\mathbf{h}}_u$, $\mathbf{G}(\overline{\nu}_u) \triangleq \mathbf{D}(\overline{\nu}_u)^{\text{H}} \mathbf{V} \in \mathbb{C}^{K \times \tilde{M}}$, $\mathbf{G}_{u'}(\overline{\nu}_u) \triangleq \mathbf{G}(\overline{\nu}_u) \mathbf{N}_{u'}^{\text{T}} \in \mathbb{C}^{K \times B_{u'}}$ and $\mathbf{N}_u \in \mathbb{R}^{B_u \times \tilde{M}}$ is the beam selection matrix with

$$[\mathbf{N}_u]_{i,j} \triangleq \begin{cases} 1, & j = [\mathcal{B}_u]_i \\ 0, & \text{else.} \end{cases} \tag{30}$$

In (29), the computation of $\overline{\mathbf{h}}_{u'}(\overline{\nu}_u)$ requires the knowledge of $\mathbf{G}(\overline{\nu}_u)$. Note that the beam matrix $\mathbf{V}$ can be viewed as chirp z-transform (CZT) matrix, hence we define $Z \triangleq e^{-\bar{\jmath} 4\pi (f_c + k\Delta f)\Delta\tau / \hat{M}}$, $\hat{M} \geq M + \tilde{M} - 1$ and

$$\boldsymbol{\xi} \triangleq \left[1, Z^{1(1-\tilde{M})/2}, \ldots, Z^{(M-1)(M-1-\tilde{M})/2} \right]^{\text{T}} \in \mathbb{C}^M, \tag{31}$$

$$\boldsymbol{\gamma} \triangleq \left[1, Z^{1^2/2}, \ldots, Z^{(\tilde{M}-1)^2/2} \right]^{\text{T}} \in \mathbb{C}^{\tilde{M}}, \tag{32}$$

$$\boldsymbol{\kappa} \triangleq \left[1, Z^{-1^2/2}, \ldots, Z^{-(M-1)^2/2} \right]^{\text{T}} \in \mathbb{C}^M, \tag{33}$$

$$\boldsymbol{\varpi} \triangleq \left[Z^{-(1-\tilde{M})^2/2}, Z^{-(2-\tilde{M})^2/2}, \cdots, Z^{-1^2/2} \right]^{\text{T}} \in \mathbb{C}^{\tilde{M}-1}, \tag{34}$$

$$\boldsymbol{\psi} \triangleq \frac{1}{\hat{M}} \mathbf{F}_{\hat{M}} \text{vec} \left\{ \left[\boldsymbol{\kappa}, \mathbf{0}_{\hat{M}-M-\tilde{M}+1}, \boldsymbol{\varpi} \right] \right\} \in \mathbb{C}^{\hat{M}}, \tag{35}$$

where $\mathbf{F}_M$ denotes the M-dimensional discrete Fourier transform (DFT) matrix, i.e., $[\mathbf{F}_M]_{i,j} = e^{-\bar{\jmath} 2\pi (i-1)(j-1)/M}$, and $\mathbf{F}_{M \times N}$ represents the matrix composed of the first N columns of $\mathbf{F}_M$. Consequently, $\mathbf{G}(\overline{\nu}_u)$ can be rewritten as

$$\mathbf{G}(\overline{\nu}_u) = \mathbf{D}(\overline{\nu}_u)^{\text{H}} \text{diag}\{\boldsymbol{\xi}\} \mathbf{F}_{\hat{M} \times M}^{\text{H}} \text{diag}\{\boldsymbol{\psi}\} \mathbf{F}_{\hat{M} \times \tilde{M}} \text{diag}\{\boldsymbol{\gamma}\}, \tag{36}$$

which can be efficiently implemented by fast Fourier transform (FFT). According to (11), $\mathcal{L}_u$ changes only when statistical CSI $\tilde{\boldsymbol{\omega}}_u$ changes. Thanks to the fact that $\tilde{\boldsymbol{\omega}}_u$ varies much more slowly than instantaneous CSI $\tilde{\mathbf{h}}_u$, thus $\mathbf{D}(\overline{\nu}_u)$ and $\mathbf{G}(\overline{\nu}_u)$ can be precalculated for each channel realization.

In conclusion, we refer to the detector discussed above as the separate Slepian transform (SST) based detector since the Slepian transform of $\mathbf{y}$ is performed for each UT separately. Algorithm 1 summarises the design procedure for the SST based detector.

Algorithm 1: SST Based Detector Design.

Input: $\hat{\mathbf{h}}_u, \rho, \mathbf{D}(\overline{\nu}_u), \mathbf{G}(\overline{\nu}_u), \mathbf{N}_u, \overline{\mathcal{U}}_u, \quad \forall u \in \mathcal{U}$
Output: $\mathbf{R}^{\text{sst}}$
1: Compute $\overline{\mathbf{h}}_{u'}(\overline{\nu}_u)$ $(\forall u' \in \overline{\mathcal{U}}_u, u \in \mathcal{U})$ by (29).
2: Compute $\mathbf{r}_1^{\text{sst}}, \ldots, \mathbf{r}_U^{\text{sst}}$ by (27).
3: Compute $\mathbf{R}^{\text{sst}}$ by (21).

D. Joint Slepian Transform Based Detector

In the previous subsection, the modulated Slepian sequences are designed for each UT separately and Slepian transforms are performed accordingly, which is relatively cumbersome and challenging to implement when UTs number becomes large. For this reason, a fixed set of modulated Slepian sequences is introduced for detection in this subsection.

Construct $\mathbf{D}(\tilde{\nu}_1), \ldots, \mathbf{D}(\tilde{\nu}_L)$ (M is divisible by L) which are independent of specific channel realizations. For the sake of simplicity, $\tilde{\nu}_1, \ldots, \tilde{\nu}_L$ are set at equal intervals on $[-1/2, 1/2)$ and symmetry about the origin, i.e.,

$$\tilde{\nu}_l = \frac{1}{L} \left(l - \frac{1}{2} \right) - \frac{1}{2}. \tag{37}$$

To ensure the detection performance, Theorem 1 implies that the following condition should be fulfilled,

$$\forall u \in \mathcal{U}, \exists l_u \in \{1, \ldots, L\}, \text{ s.t. } \mathcal{L}_u \subseteq \mathcal{M}(\tilde{\nu}_{l_u}). \tag{38}$$

Therefore, we let the length of the interval $\mathcal{M}(\tilde{\nu}_l) \cap \mathcal{M}(\tilde{\nu}_{l+1})$ not less than the maximum length of $\mathcal{L}_1, \ldots, \mathcal{L}_U$, i.e., $2W - 1/L \geq 2\overline{W}$, and select l_u by means of determining the nearest $\tilde{\nu}_l$ to $\overline{\nu}_u$. To this end, we set

$$W \geq \overline{W} + \frac{1}{2L}, \tag{39}$$

$$l_u \triangleq \arg\min_{l \in \{1, \ldots, L\}} |\tilde{\nu}_l - \overline{\nu}_u| = \lfloor L \left(\overline{\nu}_u + \frac{1}{2} \right) \rfloor + 1. \tag{40}$$

It can be proved that (38) is satisfied, the proof is provided in Appendix C. Remarkably, when W is determined, L needs to be no less than $\frac{1}{2(W - \overline{W})}$ according to (39). Nevertheless, larger L leads to higher complexity, which will be illustrated later. Hence L should be set as small as possible. Taking all these factors into account, L can be selected as the minimum integer that greater than $\frac{1}{2(W - \overline{W})}$ and can divide M at the same time, i.e.,

$$L = \min_{L' \geq \frac{1}{2(W - \overline{W})}, L' | M} L', \tag{41}$$

where $L' | M$ represents M is divisible by L. Under these conditions, K is set by (26) for the same reason.

Applying the Slepian transform to $\mathbf{y}$ after determining $\mathbf{D}(\tilde{\nu}_l)$ yields the l-th compressed observation vector

$$\begin{aligned} \overline{\mathbf{y}}(\tilde{\nu}_l) &= \sum_{u \in \tilde{\mathcal{U}}_l} \overline{\mathbf{h}}_u(\tilde{\nu}_l) x_u + \sum_{u \in \mathcal{U} \setminus \tilde{\mathcal{U}}_l} \overline{\mathbf{h}}_u(\tilde{\nu}_l) x_u + \overline{\mathbf{z}}(\tilde{\nu}_l) \\ &\overset{\text{(a)}}{\approx} \sum_{u \in \tilde{\mathcal{U}}_l} \overline{\mathbf{h}}_u(\tilde{\nu}_l) x_u + \overline{\mathbf{z}}(\tilde{\nu}_l), \end{aligned} \tag{42}$$

Authorized licensed use limited to: Southeast University. Downloaded on January 18,2024 at 06:05:40 UTC from IEEE Xplore. Restrictions apply.

Algorithm 2: JST Based Detector Design.

Input:
$\hat{\mathbf{h}}_u, \rho, \mathbf{D}(\tilde{\nu}_l), \mathbf{G}(\tilde{\nu}_l), \mathbf{N}_u, \tilde{\mathcal{U}}_l, \quad \forall u \in \mathcal{U}, l = 1, \ldots, L$
Output: $\mathbf{r}_u^{\text{jst}}, \quad \forall u \in \mathcal{U}$
1: Compute $\overline{\mathbf{h}}_u(\tilde{\nu}_l)$ $(u \in \tilde{\mathcal{U}}_l, l = 1 \cdots, L)$ by (46).
2: Compute $\mathbf{C}_1, \ldots, \mathbf{C}_L$ by (45).
3: Compute $\mathbf{r}_1^{\text{jst}}, \ldots, \mathbf{r}_U^{\text{jst}}$ by (44).

where

$$\tilde{\mathcal{U}}_l \triangleq \left\{u \middle| \mathcal{M}(\tilde{\nu}_l) \cap \mathcal{L}_u \neq \emptyset,\ u \in \mathcal{U}\right\} \text{ with } \tilde{U}_l = |\tilde{\mathcal{U}}_l|, \quad (43)$$

and (a) follows from Theorem 2.

With the compressed observation vector and the compressed channel vector in (42), the detector of UT u is given by

$$\mathbf{r}_u^{\text{jst}} = \arg\min_{\mathbf{r}_u \in \mathbb{C}^K} \mathbb{E}\left\{\left|x_u - \mathbf{r}_u^{\text{H}} \overline{\mathbf{y}}(\tilde{\nu}_{l_u})\right|^2\right\} = \mathbf{C}_{l_u}^{-1} \overline{\mathbf{h}}_u(\tilde{\nu}_{l_u}) \in \mathbb{C}^K, \quad (44)$$

where

$$\mathbf{C}_l = \sum_{u \in \tilde{\mathcal{U}}_l} \overline{\mathbf{h}}_u(\tilde{\nu}_l) \overline{\mathbf{h}}_u(\tilde{\nu}_l)^{\text{H}} + \frac{1}{\rho} \mathbf{I}_K \in \mathbb{C}^{K \times K}. \quad (45)$$

The computation of $\overline{\mathbf{h}}_u(\tilde{\nu}_l)$ involved in (44) and (45) can also be expressed as

$$\overline{\mathbf{h}}_u(\tilde{\nu}_l) = \mathbf{G}_u(\tilde{\nu}_l) \hat{\mathbf{h}}_u = \mathbf{G}(\tilde{\nu}_l) \mathbf{N}_u^{\text{T}} \hat{\mathbf{h}}_u, \quad (46)$$

where $\mathbf{G}(\tilde{\nu}_l) = \mathbf{D}(\tilde{\nu}_l)^{\text{H}} \mathbf{V}$ is fixed and independent of channel statistics, hence can be computed offline.

From another perspective, the UT u can be considered as being classified into group l_u, and all UTs are allocated to L UT sets $\mathcal{N}_1, \ldots, \mathcal{N}_L$, where

$$\mathcal{N}_l = \left\{u \middle| l_u = l,\ u \in \mathcal{U}\right\}, \quad (47)$$

represents the UT set of the l-th group, and UT u is allocated to $\mathcal{N}_{l_u}$. It can be checked that the following conditions are fulfilled.

- $\mathcal{N}_1 \cup \cdots \cup \mathcal{N}_L = \mathcal{U}$, which implies that each UT can be allocated to at least one group.
- $\mathcal{N}_l \cap \mathcal{N}_{l'} = \emptyset$ for $l \neq l'$, which indicates that each UT can only be allocated to one group.

For UTs in $\mathcal{N}_l$, the Slepian transforms of channel vectors $\overline{\mathbf{h}}_u(\tilde{\nu}_l)$, $u \in \tilde{\mathcal{U}}_l$, and the observation vector $\overline{\mathbf{y}}(\tilde{\nu}_l)$ are performed jointly, as illustrated in (42). Therefore, we refer to the detector as the joint Slepian transform (JST) based detector.

Moreover, the JST based detector can be regarded as a special case of the SST based detector by setting $\mathbf{D}(\overline{\nu}_u) = \mathbf{D}(\tilde{\nu}_{l_u})$, which means UTs in $\mathcal{N}_l$ reuse $\mathbf{D}(\tilde{\nu}_l)$ for Slepian transform. The design procedure of the JST based detector is summarized in Algorithm 2.

E. Efficient Implementation of Joint Slepian Transform Based Detection

Algorithms 1 and 2 exhibit the design procedures of proposed detectors, while the detection of transmit signals is not included. Specifically, for the JST based detector, the detection process can be efficiently implemented.

Algorithm 3: JST Based Detection.

Input: $\mathbf{y}, \mathbf{r}_u^{\text{jst}}, \mathbf{D}(\tilde{\nu}_1), \quad \forall u \in \mathcal{U}$
Output: $\hat{\mathbf{x}}^{\text{jst}}$
1: Compute $\boldsymbol{\zeta}_1, \ldots, \boldsymbol{\zeta}_K$ by (52).
2: Compute $\overline{\mathbf{y}}(\tilde{\nu}_1), \ldots, \overline{\mathbf{y}}(\tilde{\nu}_L)$ by (51).
3: Compute $\hat{\mathbf{x}}^{\text{jst}}$ by (48).

Armed with designed detector in Algorithm 2, the detection result is presented as

$$\hat{\mathbf{x}}^{\text{jst}} = \left[\left(\mathbf{r}_1^{\text{jst}}\right)^{\text{H}} \overline{\mathbf{y}}(\tilde{\nu}_{l_1}), \ldots, \left(\mathbf{r}_U^{\text{jst}}\right)^{\text{H}} \overline{\mathbf{y}}(\tilde{\nu}_{l_U})\right]^{\text{T}}. \quad (48)$$

Note that the detection process necessitates the calculation of $\overline{\mathbf{y}}(\tilde{\nu}_l)$ $(l = 1, \ldots, L)$, which can be effectively implemented by utilizing the structural properties of $\mathbf{D}(\tilde{\nu}_l)$. Construct

$$\begin{aligned}\mathbf{Q} &\triangleq [\mathbf{D}(\tilde{\nu}_1), \ldots, \mathbf{D}(\tilde{\nu}_L)] \\ &= [\text{diag}\{\mathbf{d}_1(\tilde{\nu}_1)\}, \cdots, \text{diag}\{\mathbf{d}_K(\tilde{\nu}_1)\}] \left(\mathbf{I}_K \otimes \boldsymbol{\Omega}\right) \mathbf{P}^{\text{T}}, \end{aligned} \quad (49)$$

where $\mathbf{P} \in \mathbb{R}^{KL \times KL}$ is the permutation matrix whose i-th column is $\mathbf{e}_{KL, K\langle i-1\rangle_L + \lfloor (i-1)/L \rfloor + 1}$, and

$$\boldsymbol{\Omega} \triangleq [\boldsymbol{\omega}(0), \boldsymbol{\omega}(1/L), \cdots, \boldsymbol{\omega}((L-1)/L)] = \mathbf{1}_{M/L} \otimes \mathbf{F}_L^{\text{H}}. \quad (50)$$

It can be deduced that

$$\begin{aligned}\overline{\mathbf{y}}(\tilde{\nu}_l) &= \left(\mathbf{e}_{L,l}^{\text{T}} \otimes \mathbf{I}_K\right) \text{vec}\left\{[\overline{\mathbf{y}}(\tilde{\nu}_1), \ldots, \overline{\mathbf{y}}(\tilde{\nu}_L)]\right\} \\ &= \left(\mathbf{e}_{L,l}^{\text{T}} \otimes \mathbf{I}_K\right) \mathbf{Q}^{\text{H}} \mathbf{y} \\ &= \left(\mathbf{e}_{L,l}^{\text{T}} \otimes \mathbf{I}_K\right) \mathbf{P} \left(\mathbf{I}_K \otimes \mathbf{1}_{M/L}^{\text{T}} \otimes \mathbf{F}_L\right) \\ &\quad \times \left[\text{diag}\left\{\mathbf{d}_1\left(\tilde{\nu}_1\right)\right\}, \ldots, \text{diag}\left\{\mathbf{d}_K\left(\tilde{\nu}_1\right)\right\}\right]^{\text{H}} \mathbf{y} \\ &= \left(\mathbf{e}_{L,l}^{\text{T}} \otimes \mathbf{I}_K\right) \mathbf{P} \text{vec}\left\{\mathbf{F}_L\left[\boldsymbol{\zeta}_1, \ldots, \boldsymbol{\zeta}_K\right]\right\}, \end{aligned} \quad (51)$$

where

$$\begin{aligned}\boldsymbol{\zeta}_i &\triangleq \left(\mathbf{1}_{M/L}^{\text{T}} \otimes \mathbf{I}_L\right) \left(\mathbf{d}_i(\tilde{\nu}_1)^* \odot \mathbf{y}\right) \\ &= \sum_{r=1}^{M/L} \left(\mathbf{e}_{M/L,r}^{\text{T}} \otimes \mathbf{I}_L\right) \left(\mathbf{d}_i(\tilde{\nu}_1)^* \odot \mathbf{y}\right). \end{aligned} \quad (52)$$

Consequently, the computation of $\overline{\mathbf{y}}(\tilde{\nu}_l)$ can be fast implemented by L-dimensional FFT. The process of computing $\overline{\mathbf{y}}(\tilde{\nu}_l)$ is illustrated in Fig. 2, and the detection process with the JST based detector is summarized in Algorithm 3.

F. Complexity Analysis

We discuss the computational complexity of proposed detectors in terms of required complex multiplications. To this end, define the average value of B_u, $\overline{U}_u$ and $\tilde{U}_l$ as

$$B \triangleq \frac{1}{U} \sum_{u \in \mathcal{U}} B_u, \quad \overline{U} \triangleq \frac{1}{U} \sum_{u \in \mathcal{U}} \overline{U}_u, \quad \tilde{U} \triangleq \frac{1}{L} \sum_{l=1}^{L} \tilde{U}_l, \quad (53)$$

respectively. Table I summarizes the theoretical bounds of the design complexity and the detection complexity of different

Authorized licensed use limited to: Southeast University. Downloaded on January 18,2024 at 06:05:40 UTC from IEEE Xplore. Restrictions apply.

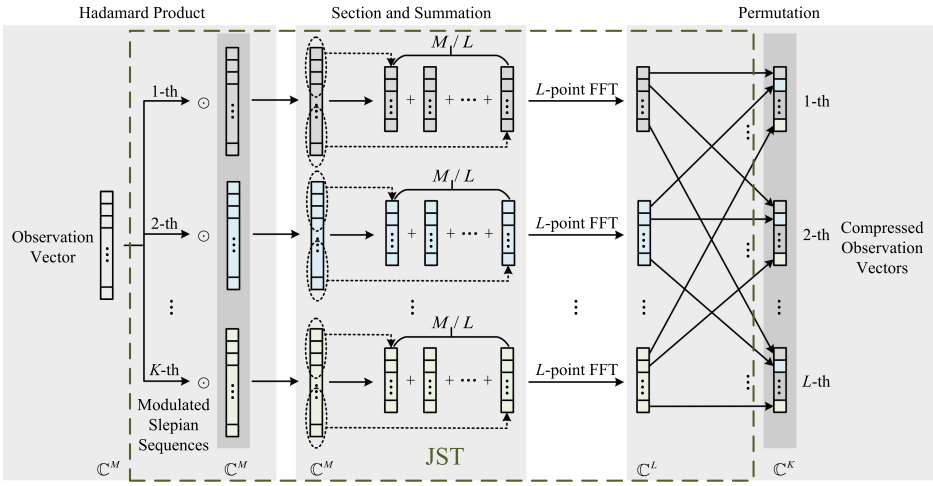


Fig. 2. Fast implementation of computing $\overline{\mathbf{y}}(\tilde{\nu}_l)$.

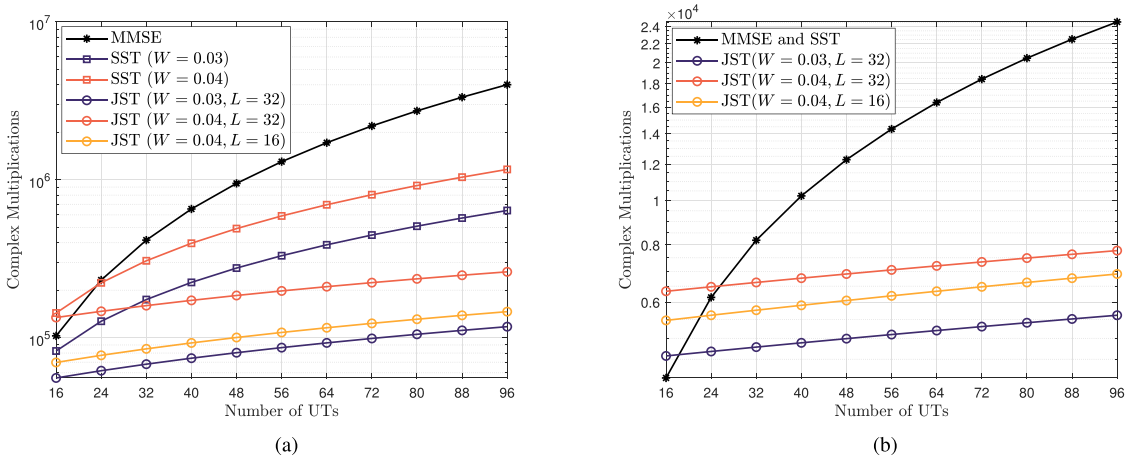


Fig. 3. Complexity comparison against UTs number (from 16 to 96). Parameters are set as $M = 256$, $\lambda^{\text{th}} = 0.99$ ($K = \lfloor 2MW \rfloor - 2$). (a) Design complexity. (b) Detection complexity.

TABLE I
COMPUTATIONAL COMPLEXITY OF EACH DETECTOR

Detector	Design complexity	Detection complexity
MMSE	$\frac{MU(U+1)}{2}+\frac{U^3}{2}+\frac{3U^2}{2}+MU^2$	MU
SST	$\left(\left(B+\frac{(K+1)}{2}\right)\overline{U}+\frac{K^2}{2}+\frac{5K}{2}+M\right)KU$	MU
JST	$\left(\left(B+\frac{(K+1)}{2}\right)\tilde{U}+\frac{K^2}{2}+\frac{3K}{2}\right)KL+K^2U$	$K(M+U+\frac{L}{2}\log L)$

detectors, respectively. Fig. 3 plots the results under the given system configuration with different parameter settings.

We first focus on the design complexity of the detector. For the SST based detector, this term comes from the computation of (21), (27) and (29). It is worth pointing out that in (29), by exploiting the sparsity of the beam domain channel, the complex multiplications required for computing $\overline{\mathbf{h}}_{u'}(\overline{\nu}_u)$ are reduced from KM to KB_u, and only $\overline{U}_u$ rather than U compressed channel vectors need to be computed for each UT u, which is crucial for complexity reduction. Specifically, (21) involves the computation of U matrix-vector multiplications with complexity MKU. Furthermore, (27) and (29) involve $U(\overline{U}+1)$ matrix-vector multiplications, the computation of U Gram matrices, and inversions of U positive definite matrices, whose complexities are given by $BK\overline{U}U + K^2U$, $0.5(K+1)K\overline{U}U$ and $(0.5K^3+1.5K^2)U$ [41], respectively. Similarly, for the JST based detector, the design complexity comes from the computation of (44), (45) and (46), which is $((B+0.5(K+1))\tilde{U}+0.5K^2+1.5K)KL+K^2U$ in total. When UTs number is large, the complexity of the JST based detector is dramatically reduced compared to the SST based detector since it performs a fixed set of the Slepian transforms instead of performing for each UT separately, which is insensitive to UTs number variations. As baseline, the design complexity of the MMSE detector is $0.5MU(U+1)+0.5U^3+1.5U^2+MU^2$. As shown in Fig. 3(a), when UTs number is small, the design complexities of proposed detectors are comparable to that of the MMSE detector. However, when UTs number gets larger, the advantage of design complexity for the proposed detectors becomes rather apparent, especially for the JST based one.

Authorized licensed use limited to: Southeast University. Downloaded on January 18,2024 at 06:05:40 UTC from IEEE Xplore. Restrictions apply.

TABLE II
SIMULATION PARAMETERS

Parameter	Value
Carrier frequency f_c	16 MHz
Number of antennas at BS M	256
Subcarriers spacing Δf	250 Hz
Antenna spacing at BS d	9 m
Number of subcarriers N_c	2048
Number of valid subcarriers N_v	1536
Modulation scheme	16QAM
Number of UTs U	96

Next, we discuss the detection complexity with a designed detector. For the SST based detector, this term comes from the computation of (20), which is equal to that of the MMSE detector, i.e., MU. Different from the SST based detector, for the JST based detector, $\overline{\mathbf{y}}(\tilde{\nu}_l)$ is computed for complexity reduction. Specifically, the detection complexity comes from the computation of (48), (51) and (52), involving U inner products of K-dimensional vectors, K Hadamard products of M-dimensional vectors, and K L-point FFT, which is $K(M+U+0.5L\log L)$ in total. From Fig. 3(b), it can be observed that when UTs number is small, due to the dominant computational part $\mathbf{d}_i(\tilde{\nu}_1)^* \odot \mathbf{y}$, the detection complexity of the JST based detector is slightly higher than that of the MMSE detector, and its complexity advantage is only highlighted when the UTs number is sufficiently large. Remarkably, the computational cost of the JST based detector is suppressed below 10^4 complex multiplications even in $(M,U,W,L)=(256,96,0.04,32)$ configuration, about one-third of that of the MMSE detector.

IV. SIMULATION RESULTS

In this section, simulation results are provided to show the performance of the proposed detectors for HF skywave massive MIMO-OFDM systems.

A. Simulation Setup

In the simulation, we consider HF skywave massive MIMO-OFDM systems and the main parameters are given in Table II.

We utilize Proplab-Pro version 3.1 [42], a commercial ray-tracing software, to generate the realistic HF skywave channel parameters. The location of the BS is set as (34°N, 118°E). UTs are generated at a distance of 2000 km from the BS, and the azimuth angles of UTs seen from the BS are generated in the interval $[-70°, 70°]$. We perform the user scheduling to select UTs with relatively low channel correlation. The ray-tracing process is performed at 11:00 coordinated universal time on March 10, 2021. Isotropic antennas are assumed for BS and UTs. The signal strength, propagation distance, azimuth AoD and elevation AoD for each path are provided by ray-tracing. Thus, we can calculate the path gain, propagation delay and direction cosine for each path accordingly. In addition, the initial phases and Doppler shifts for paths are generated randomly. Consequently, the space domain channel is generated according to (4). The corresponding beam domain channel is acquired by the method proposed in [33] with $\tilde{M}=2M$. The noise power in a one Hz bandwidth is set as -163 dBW at the BS according to the voice of America coverage analysis program (VOACAP) [6], [43].

TABLE III
EIGENVALUES OF $\mathbf{B}$

K	$\lfloor 2MW \rfloor - 4$	$\lfloor 2MW \rfloor - 3$	$\lfloor 2MW \rfloor - 2$
λ_K	0.999883244560431	0.998993962635411	0.992604155756452
K	$\lfloor 2MW \rfloor - 1$	$\lfloor 2MW \rfloor$	$\lfloor 2MW \rfloor + 1$
λ_K	0.955864536301262	0.809672713861650	0.492316172731189
K	$\lfloor 2MW \rfloor + 2$	$\lfloor 2MW \rfloor + 3$	$\lfloor 2MW \rfloor + 4$
λ_K	0.180853364150025	0.041637897002406	0.007059649484042

B. Detection Performance

To evaluate the performance of proposed detectors, we discuss the bit error rate (BER) versus the total transmit power ϵ (in dBW) without channel coding, and hard decision is utilized. Since proposed detectors are derived from the MMSE criterion, and from (22) we know that the MMSE detector provides a mean-squared error lower bound for proposed detectors, thus it is reasonable to introduce the performance of the MMSE detector as a reference.

In Fig. 4, the detection performance of proposed detectors with different values of W are compared. From the figure, the detection performances for both the SST based detector and the JST based detector improve as W increases. When W is large, the detection performance of the JST based detector is almost the same as that of the SST based detector, and are closely approaching that of the MMSE detector. Moreover, when W is small, the detection performances of proposed detectors are far inferior to that of the MMSE detector, and as a result of the unsatisfaction of $\mathcal{L}_u \subseteq \mathcal{M}(\tilde{\nu}_{l_u})$, the JST based detector has a more significant gap than that of the SST based detector. However, the complexities of proposed detectors increase as W increases since the effect of dimensionality reduction is not remarkable when W is large. Therefore, W should be as large as possible with acceptable detection complexity.

Fig. 5 compares the detection performances of proposed detectors with different values of K, where original SST (JST) stands for the detector without simplification, i.e., all inter-UT interferences are considered. From the figure, for original ones, the detection performances of proposed detectors improve as K increases. However, for simplified ones, when K increases, the performances of proposed detectors first improve and then decrease. To interpret this result, we list the eigenvalues of $\mathbf{B}$ corresponding to different K as in Table III. When $K < \lfloor 2MW \rfloor - 2$, there is a large gap between K and $2MW$, then in the Slepian transform process, the channel energy of UTs is heavily lost, which brings about the degradation of the detection performance. When $K > \lfloor 2MW \rfloor - 2$, the gap between λ_K and 1 implies there is remarkable spectral diffusion outside the interval $\mathcal{M}(\overline{\nu}_u)$ $(\mathcal{M}(\tilde{\nu}_{l_u}))$ for $\mathbf{d}_K(\overline{\nu}_u)$ $(\mathbf{d}_K(\tilde{\nu}_{l_u}))$, which aggravates the inter-UT interference, thereby the simplification of proposed detectors also leads to degraded detection performance, especially at high transmit power region. Therefore,

Authorized licensed use limited to: Southeast University. Downloaded on January 18,2024 at 06:05:40 UTC from IEEE Xplore. Restrictions apply.

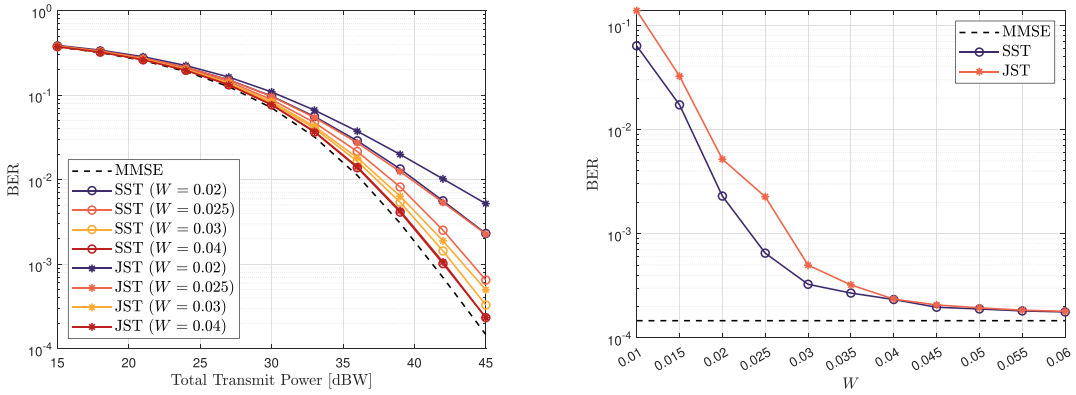


Fig. 4. BER performance comparison against (a) the total transmit power ϵ (b) W ($\epsilon = 45$ dBW), and (λ^{th}, L) = (0.99, 32).

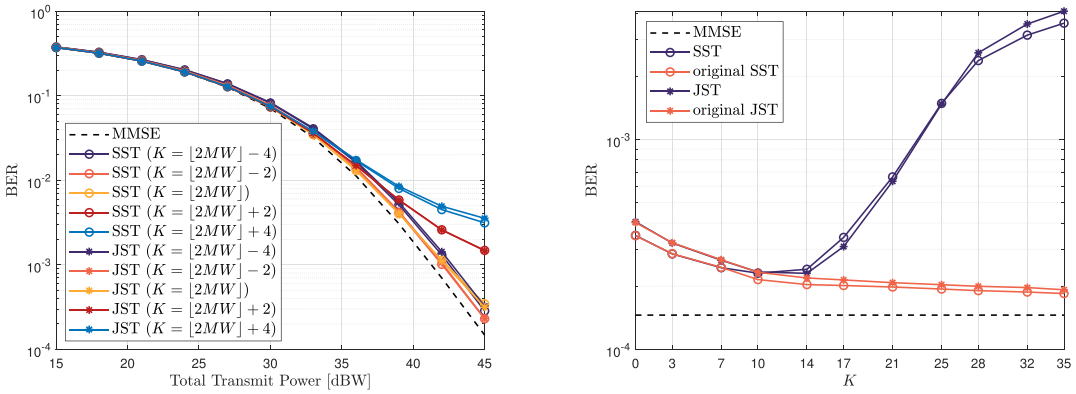


Fig. 5. BER performance comparison against (a) the total transmit power ϵ (b) K ($\epsilon = 45$ dBW), and (W, L) = (0.04, 32).

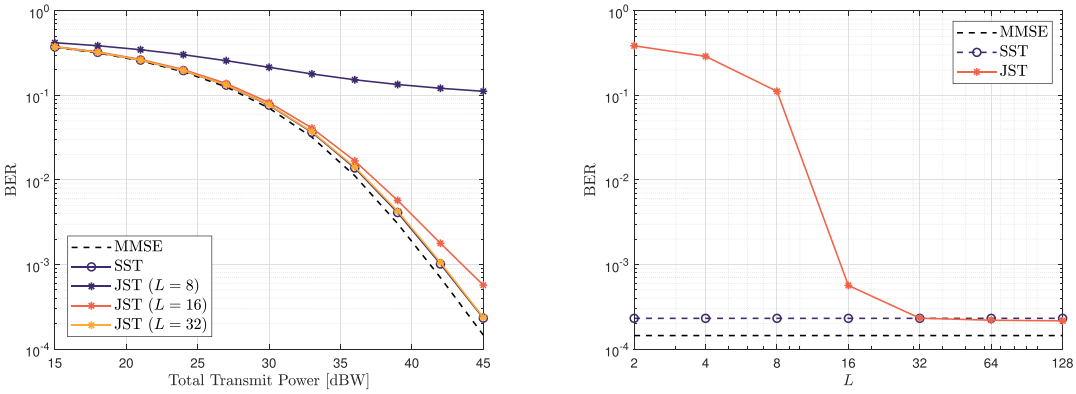


Fig. 6. BER performance comparison against (a) the total transmit power ϵ (b) L ($\epsilon = 45$ dBW), and (W, λ^{th}) = (0.04, 0.99).

$K = \lfloor 2MW \rfloor - 2$ is the best selection in this scenario and $\lambda^{\text{th}} = 0.99$ is a reasonable threshold.

Next, we compare the detection performance of the JST based detector with various values of L, including the results of the SST based detector and the MMSE detector as baselines, and curves are plotted in Fig. 6. According to the complexity analysis, we know that smaller L results in lower complexity, whereas the detection performance deteriorates with L decreases, as demonstrated in the figure. Specifically, $L \leq 8$ yields extremely unacceptable detection performance as (39) is not met. Consequently, the trade-off between detection performance and complexity can be balanced with $L = 32$.

To demonstrate the effectiveness of proposed detectors, we set the ratio $R = M/U$ close to 1, in which the number of UTs is set as $U = 208$. The detection performances of proposed detectors with different parameter settings are given in Fig. 7. It can be observed from the figure that when R is close to 1, larger W is required, since extensive inter-UT interference

Authorized licensed use limited to: Southeast University. Downloaded on January 18,2024 at 06:05:40 UTC from IEEE Xplore. Restrictions apply.

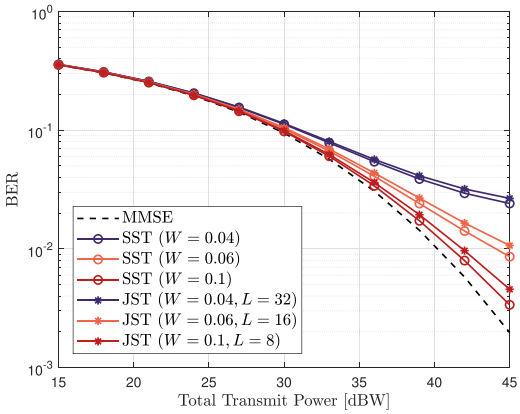


Fig. 7. BER performance comparsion against the total transmit power with $U = 208$ and $K = \lfloor 2MW \rfloor - 2$.

occurs. Specifically, when $W = 0.1$, $L = 8$, proposed detectors can achieve desirable performance. In this case the SST based detector has no complexity advantage since U is too large. Nevertheless, the number of multiplications for the design and the detection of the JST based detector are 1702671 and 23324, respectively, while those of the MMSE detector are 21204352 and 53248, respectively, which shows that the JST based detector still exhibits low complexity in this scenario. Therefore, in terms of both the detection performance and the complexity, the SST based detector can be selected when R is relatively large, whereas the JST based detector is prefered when R is relatively small.

V. Conclusion

In this article, we have investigated detectors based on Slepian transform for HF skywave massive MIMO-OFDM systems. We have introduced a wideband beam based massive MIMO-OFDM channel model for HF skywave communications. By exploiting beam domain sparsity of the channel model, we designed modulated Slepian sequences for each UT and performed Slepian transform on the observation vector and the channel matrix for dimensionality reduction. Then the SST based detector is proposed, and the design complexity of the detector is lower than that of the MMSE detector, facilitated by dimensionality reduction and the utilization of the sparse beam domain channel. To further reduce the complexity of the SST based detector, we developed the JST based detector based on fixed modulated Slepian sequences. Due to the fixed number of Slepian transforms and the implementation of FFT, when the antenna number and UTs number are both large, the complexity of the JST based detector is significantly lower than the MMSE and the SST based detector. The simulation results show that by choosing appropriate parameters, proposed detectors can provide similar performance to that of the MMSE detector at a much lower complexity. For future research, we can apply the Slepian transform approach to nonlinear detection methods and downlink (DL) precoding problems, etc.

Appendix A
Proof of Theorem 1

We introduce the following lemmas which are required in the proof of Theorem 1.

Lemma 1: When $\mathcal{P}_u \bigcap \mathcal{P}_{u'} = \emptyset$ for $u \neq u'$, we have

$$\lim_{M\to\infty} (\mathbf{h}_u C_u - \mathbf{C}\mathbf{h}_u) = \mathbf{0}_M, \tag{54}$$

where $C_u \triangleq (\mathbf{h}_u^{\mathrm{H}}\mathbf{h}_u + \frac{1}{\rho})^{-1}$, $\mathbf{C} \triangleq (\mathbf{H}\mathbf{H}^{\mathrm{H}} + \frac{1}{\rho}\mathbf{I}_M)^{-1}$.

Proof: According to the arithmetic of limits,

$$\begin{aligned}
&\lim_{M\to\infty} (\mathbf{h}_u C_u - \mathbf{C}\mathbf{h}_u) \\
&= \lim_{M\to\infty} \frac{1}{M}\mathbf{C}\left(\mathbf{C}^{-1}\mathbf{h}_u - \mathbf{h}_u C_u^{-1}\right) \lim_{M\to\infty} MC_u \\
&= \left(\sum_{u'\neq u} \lim_{M\to\infty} \frac{1}{M}\mathbf{C}\mathbf{h}_{u'}\mathbf{h}_{u'}^{\mathrm{H}}\mathbf{h}_u\right) \lim_{M\to\infty} MC_u.
\end{aligned} \tag{55}$$

When $\mathcal{P}_u \cap \mathcal{P}_{u'} = \emptyset$ for $u \neq u'$, according to [6, Lemma 1], we have

$$\lim_{M\to\infty} \frac{1}{M}\mathbf{h}_{u'}^{\mathrm{H}}\mathbf{h}_u = \delta[u-u'] \sum_{p=1}^{P_u} \beta_{u,p}^2. \tag{56}$$

Substituting (56) into (55),

$$\lim_{M\to\infty} (\mathbf{h}_u C_u - \mathbf{C}\mathbf{h}_u) = \mathbf{0}_M \cdot \left(\sum_{p=1}^{P_u} \beta_{u,p}^2\right)^{-1} = \mathbf{0}_M. \tag{57}$$

This completes the proof of Lemma 1.

Lemma 2: For $i > K$, we have

$$\|\overline{\mathbf{C}}_u \mathbf{d}_i(\overline{\nu}_u)\| \leq \rho\sqrt{1+\rho M E}, \tag{58}$$

where $E \triangleq \sum_{u\in\mathcal{U}} E_u$ with $E_u \triangleq (\sum_{p=1}^{P_u} \beta_{u,p})^2$, and $\overline{\mathbf{C}}_u \triangleq (\mathbf{D}(\overline{\nu}_u)\mathbf{D}(\overline{\nu}_u)^{\mathrm{H}}\mathbf{H}\mathbf{H}^{\mathrm{H}} + \frac{1}{\rho}\mathbf{I}_M)^{-1}$.

Proof: When $i > K$, $\mathbf{d}_i(\overline{\nu}_u)^{\mathrm{H}}\mathbf{D}(\overline{\nu}_u) = \mathbf{0}_K^{\mathrm{T}}$, then we have

$$\mathbf{d}_i(\overline{\nu}_u)^{\mathrm{H}}\overline{\mathbf{C}}_u^{-1} = \frac{1}{\rho}\mathbf{d}_i(\overline{\nu}_u)^{\mathrm{H}} \Rightarrow \mathbf{d}_i(\overline{\nu}_u)^{\mathrm{H}}\overline{\mathbf{C}}_u = \rho\mathbf{d}_i(\overline{\nu}_u)^{\mathrm{H}}, \tag{59}$$

which yields

$$\begin{aligned}
\rho &= \rho\mathbf{d}_i(\overline{\nu}_u)^{\mathrm{H}}\mathbf{d}_i(\overline{\nu}_u) \\
&= \mathbf{d}_i(\overline{\nu}_u)^{\mathrm{H}}\overline{\mathbf{C}}_u^{\mathrm{H}}\left(\overline{\mathbf{C}}_u^{\mathrm{H}}\right)^{-1}\overline{\mathbf{C}}_u\mathbf{d}_i(\overline{\nu}_u) \\
&= \mathbf{d}_i(\overline{\nu}_u)^{\mathrm{H}}\overline{\mathbf{C}}_u^{\mathrm{H}}\mathbf{H}\mathbf{H}^{\mathrm{H}}\mathbf{D}(\overline{\nu}_u)\mathbf{D}(\overline{\nu}_u)^{\mathrm{H}}\overline{\mathbf{C}}_u\mathbf{d}_i(\overline{\nu}_u) + \frac{1}{\rho}\|\overline{\mathbf{C}}_u\mathbf{d}_i(\overline{\nu}_u)\|^2 \\
&\overset{(a)}{=} \mathbf{d}_i(\overline{\nu}_u)^{\mathrm{H}}\mathbf{H}\hat{\mathbf{C}}_u^2\mathbf{H}^{\mathrm{H}}\mathbf{d}_i(\overline{\nu}_u) - \rho\mathbf{d}_i(\overline{\nu}_u)^{\mathrm{H}}\mathbf{H}\hat{\mathbf{C}}_u\mathbf{H}^{\mathrm{H}}\mathbf{d}_i(\overline{\nu}_u) \\
&\quad + \frac{1}{\rho}\|\overline{\mathbf{C}}_u\mathbf{d}_i(\overline{\nu}_u)\|^2,
\end{aligned} \tag{60}$$

where $\hat{\mathbf{C}}_u \triangleq (\mathbf{H}^{\mathrm{H}}\mathbf{D}(\overline{\nu}_u)\mathbf{D}(\overline{\nu}_u)^{\mathrm{H}}\mathbf{H} + \frac{1}{\rho}\mathbf{I}_U)^{-1}$, and (a) follows from the fact $\overline{\mathbf{C}}_u^{\mathrm{H}}\mathbf{H} = \mathbf{H}\hat{\mathbf{C}}_u$ and $\mathbf{D}(\overline{\nu}_u)^{\mathrm{H}}\mathbf{d}_i(\overline{\nu}_u) = \mathbf{0}_K$. Therefore,

$$\|\overline{\mathbf{C}}_u\mathbf{d}_i(\overline{\nu}_u)\|^2 = \rho^2\left(1+\mathbf{d}_i(\overline{\nu}_u)^{\mathrm{H}}\mathbf{H}\hat{\mathbf{C}}_u\mathbf{H}^{\mathrm{H}}\mathbf{d}_i(\overline{\nu}_u)\right)$$

Authorized licensed use limited to: Southeast University. Downloaded on January 18,2024 at 06:05:40 UTC from IEEE Xplore. Restrictions apply.

$$
\begin{aligned}
&-\rho \mathbf{d}_i(\overline{\nu}_u)^{\mathrm{H}}\mathbf{H}\hat{\mathbf{C}}_u^2\mathbf{H}^{\mathrm{H}}\mathbf{d}_i(\overline{\nu}_u)\\
&\le \rho^2\left(1+\mathbf{d}_i(\overline{\nu}_u)^{\mathrm{H}}\mathbf{H}\hat{\mathbf{C}}_u\mathbf{H}^{\mathrm{H}}\mathbf{d}_i(\overline{\nu}_u)\right)\\
&\overset{(b)}{\le} \rho^2\left(1+\rho\sum_{u'\in\mathcal{U}}\mathbf{d}_i(\overline{\nu}_u)^{\mathrm{H}}\mathbf{h}_{u'}\mathbf{h}_{u'}^{\mathrm{H}}\mathbf{d}_i(\overline{\nu}_u)\right)\\
&\overset{(c)}{\le} \rho^2\left(1+\rho\sum_{u'\in\mathcal{U}}\|\mathbf{d}_i(\overline{\nu}_u)\|^2\|\mathbf{h}_{u'}\|^2\right)\\
&\le \rho^2(1+\rho ME), \qquad (61)
\end{aligned}
$$

where (b) follows from [38],

$$\rho\mathbf{I}_U-\hat{\mathbf{C}}_u\succeq\hat{\mathbf{C}}_u\mathbf{H}^{\mathrm{H}}\mathbf{D}(\overline{\nu}_u)\mathbf{D}(\overline{\nu}_u)^{\mathrm{H}}\mathbf{H}\hat{\mathbf{C}}_u\succeq\mathbf{O}_{U\times U}, \qquad (62)$$

and (c) follows from the compatibility of the Frobenius norm. This completes the proof of Lemma 2.

It can be deduced that

$$
\begin{aligned}
&\mathbb{E}\left\{\|\hat{\mathbf{x}}^{\mathrm{mmse}}-\hat{\mathbf{x}}^{\mathrm{sst}}\|^2\right\}\\
&=\sum_{u\in\mathcal{U}}\mathbb{E}\left\{\mathbf{h}_u^{\mathrm{H}}\left(\mathbf{C}-\tilde{\mathbf{C}}_u\right)\mathbf{y}\mathbf{y}^{\mathrm{H}}\left(\mathbf{C}-\tilde{\mathbf{C}}_u\right)\mathbf{h}_u\right\}\\
&=\sigma_{\mathrm{x}}^2\sum_{u\in\mathcal{U}}\mathbf{h}_u^{\mathrm{H}}\left(\mathbf{C}-\tilde{\mathbf{C}}_u\right)\mathbf{C}^{-1}\left(\mathbf{C}-\tilde{\mathbf{C}}_u\right)\mathbf{h}_u\\
&=\sigma_{\mathrm{x}}^2\sum_{u\in\mathcal{U}}\mathbf{h}_u^{\mathrm{H}}\left(\mathbf{I}_M-\tilde{\mathbf{C}}_u\mathbf{C}^{-1}\right)\left(\mathbf{C}-\tilde{\mathbf{C}}_u\right)\mathbf{h}_u\\
&\overset{(a)}{=}\sigma_{\mathrm{z}}^2\sum_{u\in\mathcal{U}}\mathbf{h}_u^{\mathrm{H}}\overline{\mathbf{C}}_u\left(\mathbf{I}_M-\mathbf{D}(\overline{\nu}_u)\mathbf{D}(\overline{\nu}_u)^{\mathrm{H}}\right)\mathbf{C}\mathbf{h}_u\\
&=\sigma_{\mathrm{z}}^2\sum_{u\in\mathcal{U}}\sum_{i=K+1}^{M}\mathbf{h}_u^{\mathrm{H}}\overline{\mathbf{C}}_u\mathbf{d}_i(\overline{\nu}_u)\mathbf{d}_i(\overline{\nu}_u)^{\mathrm{H}}\mathbf{C}\mathbf{h}_u, \qquad (63)
\end{aligned}
$$

where

$$\tilde{\mathbf{C}}_u\triangleq\mathbf{D}(\overline{\nu}_u)\left(\mathbf{D}(\overline{\nu}_u)^{\mathrm{H}}\mathbf{H}\mathbf{H}^{\mathrm{H}}\mathbf{D}(\overline{\nu}_u)+\frac{1}{\rho}\mathbf{I}_K\right)^{-1}\mathbf{D}(\overline{\nu}_u)^{\mathrm{H}}, \qquad (64)$$

and (a) follows from the fact that $\tilde{\mathbf{C}}_u=\overline{\mathbf{C}}_u\mathbf{D}(\overline{\nu}_u)\mathbf{D}(\overline{\nu}_u)^{\mathrm{H}}$ and $\mathbf{D}(\overline{\nu}_u)^{\mathrm{H}}\mathbf{D}(\overline{\nu}_u)=\mathbf{I}_K$.

Then, according to Lemma 1,

$$
\begin{aligned}
&\lim_{M\to\infty}\mathbb{E}\left\{\|\hat{\mathbf{x}}^{\mathrm{mmse}}-\hat{\mathbf{x}}^{\mathrm{sst}}\|^2\right\}\\
&=\lim_{M\to\infty}\sigma_{\mathrm{z}}^2\left|\sum_{u\in\mathcal{U}}\sum_{i=K+1}^{M}\mathbf{h}_u^{\mathrm{H}}\overline{\mathbf{C}}_u\mathbf{d}_i(\overline{\nu}_u)\mathbf{d}_i(\overline{\nu}_u)^{\mathrm{H}}\mathbf{h}_uC_u\right|\\
&\overset{(b)}{\le}\lim_{M\to\infty}\sigma_{\mathrm{x}}^2\sum_{u\in\mathcal{U}}\sum_{i=K+1}^{M}\|\mathbf{h}_u\|\|\overline{\mathbf{C}}_u\mathbf{d}_i(\overline{\nu}_u)\|\left|\mathbf{d}_i(\overline{\nu}_u)^{\mathrm{H}}\mathbf{h}_u\right|, \qquad (65)
\end{aligned}
$$

where (b) follows from the compatibility of the Frobenius norm and $C_u\le\rho$.

Moreover, according to the Parseval's theorem,

$$\lim_{M\to\infty}\left|\mathbf{d}_i(\overline{\nu}_u)^{\mathrm{H}}\mathbf{h}_u\right|=\lim_{M\to\infty}\left|\int_{\mathcal{L}_u}D_i(\overline{\nu}_u,\nu)^*\tilde{h}_u(\nu)\mathrm{d}\nu\right|, \qquad (66)$$

where $D_i(\overline{\nu}_u,\nu)=\mathcal{F}\{d_i[n]\}$ with

$$d_i[n]\triangleq\begin{cases}[\mathbf{d}_i(\overline{\nu}_u)]_{n+1}, & n=0,\ldots,M-1\\ 0, & \text{else.}\end{cases} \qquad (67)$$

For $\mathcal{L}_u\subseteq\mathcal{M}(\overline{\nu}_u)$,

$$
\begin{aligned}
&\left|\int_{\mathcal{L}_u}D_i(\overline{\nu}_u,\nu)^*\tilde{h}_u(\nu)\mathrm{d}\nu\right|^2\\
&\overset{(c)}{\le}\left(\int_{\mathcal{M}(\overline{\nu}_u)}|D_i(\overline{\nu}_u,\nu)|^2\mathrm{d}\nu\right)\left(\int_{\mathcal{L}_u}|\tilde{h}_u(\nu)|^2\mathrm{d}\nu\right)\overset{(d)}{=}\lambda_i\mathcal{E}_u, \qquad (68)
\end{aligned}
$$

where (c) follows from the Cauchy-Schwarz inequality [44] and

$$
\begin{aligned}
&\int_{\mathcal{L}_u}|D_i(\overline{\nu}_u,\nu)|^2\mathrm{d}\nu\\
&\le\int_{\mathcal{L}_u+\mathcal{M}(\overline{\nu}_u)\setminus\mathcal{L}_u}|D_i(\overline{\nu}_u,\nu)|^2\mathrm{d}\nu=\int_{\mathcal{M}(\overline{\nu}_u)}|D_i(\overline{\nu}_u,\nu)|^2\mathrm{d}\nu, \qquad (69)
\end{aligned}
$$

(d) follows from

$$
\begin{aligned}
\lambda_i&=\frac{\mathbf{s}_i^{\mathrm{H}}\mathbf{B}\mathbf{s}_i}{\mathbf{s}_i^{\mathrm{H}}\mathbf{s}_i}\\
&=\frac{\mathbf{d}_i(\overline{\nu}_u)^{\mathrm{H}}\overline{\mathbf{B}}(\overline{\nu}_u)\mathbf{d}_i(\overline{\nu}_u)}{\mathbf{d}_i(\overline{\nu}_u)^{\mathrm{H}}\mathbf{d}_i(\overline{\nu}_u)}\\
&=\frac{\int_{\mathcal{M}(\overline{\nu}_u)}|D_i(\overline{\nu}_u,\nu)|^2\mathrm{d}\nu}{\int_{-\frac{1}{2}}^{\frac{1}{2}}|D_i(\overline{\nu}_u,\nu)|^2\mathrm{d}\nu}\\
&=\int_{\mathcal{M}(\overline{\nu})}|D_i(\overline{\nu}_u,\nu)|^2\mathrm{d}\nu, \qquad (70)
\end{aligned}
$$

where $\overline{\mathbf{B}}(\overline{\nu}_u)\triangleq\mathrm{diag}\{\boldsymbol{\omega}(\overline{\nu}_u)\}\mathbf{B}\mathrm{diag}\{\boldsymbol{\omega}(\overline{\nu}_u)\}^{\mathrm{H}}$ and $\mathcal{E}_u\triangleq\sum_{m\in\mathcal{A}}|h_u[m]|^2\le ME_u$. According to [18] and [45, Lemma 4.2], when $i>K=\lfloor 2MW(1+\epsilon)\rfloor$, $0<\epsilon<\frac{1}{2W}-1$ and $M\ge M_{W,\epsilon}$, we have the following inequality

$$\left|\int_{\mathcal{M}(\overline{\nu})}D_i(\overline{\nu}_u,\nu)^*\tilde{h}_u(\nu)\mathrm{d}\nu\right|\le\sqrt{\lambda_i\mathcal{E}_u}\le\sqrt{E_uMe^{-C_{W,\epsilon}M}}, \qquad (71)$$

where $C_{W,\epsilon}$ and $M_{W,\epsilon}$ are positive constants depend on W and ϵ. Substituting (66), (71) into (65), and according to Lemma 2 and the squeeze theorem, we have

$$
\begin{aligned}
&\lim_{M\to\infty}\mathbb{E}\left\{\|\hat{\mathbf{x}}^{\mathrm{mmse}}-\hat{\mathbf{x}}^{\mathrm{sst}}\|^2\right\}\\
&=\lim_{M\to\infty}\sigma_{\mathrm{x}}^2\rho\sum_{u\in\mathcal{U}}(M-K)\sqrt{ME_u}\sqrt{1+\rho ME}\sqrt{E_uMe^{-C_{W,\epsilon}M}}\\
&=\lim_{M\to\infty}\sigma_{\mathrm{x}}^2\rho E(1+\rho ME)^{1/2}M^2e^{-C_{W,\epsilon}M/2}\\
&=0. \qquad (72)
\end{aligned}
$$

This completes the proof.

Appendix B
Proof of Theorem 2

As $M\to\infty$, the inner product between $\mathbf{d}_i(\overline{\nu}_u)$ and $\mathbf{h}_{u'}$ is

$$\lim_{M\to\infty}\mathbf{d}_i(\overline{\nu}_u)^{\mathrm{H}}\mathbf{h}_{u'}=\lim_{M\to\infty}\int_{\mathcal{L}_u}D_i(\overline{\nu}_u,\nu)^*\tilde{h}_{u'}(\nu)\mathrm{d}\nu. \qquad (73)$$

Authorized licensed use limited to: Southeast University. Downloaded on January 18,2024 at 06:05:40 UTC from IEEE Xplore. Restrictions apply.

In a similar manner to (68), when $\mathcal{M}(\overline{\nu}_u) \cap \mathcal{L}_{u'} = \emptyset$,

$$\begin{aligned}
&\left| \int_{\mathcal{L}_u} D_i(\overline{\nu}_u, \nu)^* \tilde{h}_{u'}(\nu) \mathrm{d}\nu \right|^2 \\
&\leq \left(\int_{\mathcal{L}_u} |D_i(\overline{\nu}_u, \nu)|^2 \mathrm{d}\nu \right) \left(\int_{\mathcal{L}_u} |\tilde{h}_{u'}(\nu)|^2 \mathrm{d}\nu \right) \\
&\leq \left(1 - \int_{\mathcal{M}(\overline{\nu}_u)} |D_i(\overline{\nu}_u, \nu)|^2 \mathrm{d}\nu \right) \left(\int_{-\frac{1}{2}}^{\frac{1}{2}} |\tilde{h}_{u'}(\nu)|^2 \mathrm{d}\nu \right) \\
&= (1 - \lambda_i) \mathcal{E}_{u'}. \qquad (74)
\end{aligned}$$

According to [18] and [45, Lemma 4.1], when $i \leq K = \lfloor 2MW(1-\epsilon) \rfloor$, $0 < \epsilon < 1$ and $M \geq \overline{M}_{W,\epsilon}$, we have the following inequality

$$\begin{aligned}
&\left| \int_{\mathcal{L}_u} D_i(\overline{\nu}_u, \nu)^* \tilde{h}_{u'}(\nu) \mathrm{d}\nu \right| \\
&\leq \sqrt{(1 - \lambda_i) \mathcal{E}_u} \leq \sqrt{E_{u'} M e^{-\overline{C}_{W,\epsilon} M}}, \qquad (75)
\end{aligned}$$

where $\overline{C}_{W,\epsilon}$ and $\overline{M}_{W,\epsilon}$ are positive constants depend on W and ϵ. Therefore, for $i \leq K$,

$$\lim_{M \to \infty} \left| \mathbf{d}_i(\overline{\nu}_u)^{\mathrm{H}} \mathbf{h}_{u'} \right| = \lim_{M \to \infty} \sqrt{E_{u'} M e^{-\overline{C}_{W,\epsilon} M}} = 0, \qquad (76)$$

$$\Rightarrow \lim_{M \to \infty} \mathbf{D}(\overline{\nu}_u)^{\mathrm{H}} \mathbf{h}_{u'} = \mathbf{0}_K. \qquad (77)$$

This completes the proof.

Appendix C
Proof of (38)

From (37) and (40), we have

$$\begin{aligned}
L \left| \tilde{\nu}_{l_u} - \overline{\nu}_u \right| &= L \left| \frac{1}{L} \left(\lfloor L \left(\overline{\nu}_u + \frac{1}{2} \right) \rfloor + \frac{1}{2} \right) - \frac{1}{2} - \overline{\nu}_u \right| \\
&= \left| \frac{1}{2} + \left(\lfloor L \left(\overline{\nu}_u + \frac{1}{2} \right) \rfloor - L \left(\overline{\nu}_u + \frac{1}{2} \right) \right) \right|, \qquad (78)
\end{aligned}$$

note that

$$-1 < \lfloor L \left(\overline{\nu}_u + \frac{1}{2} \right) \rfloor - L \left(\overline{\nu}_u + \frac{1}{2} \right) \leq 0, \qquad (79)$$

then

$$L \left| \tilde{\nu}_{l_u} - \overline{\nu}_u \right| \leq \frac{1}{2} \Rightarrow \overline{\nu}_u - \frac{1}{2L} \leq \tilde{\nu}_{l_u} \leq \overline{\nu}_u + \frac{1}{2L}. \qquad (80)$$

Combining (39), we have

$$\nu_u^{\mathrm{l}} \geq \overline{\nu}_u - \overline{W} \geq \overline{\nu}_u + \frac{1}{2L} - W \geq \tilde{\nu}_{l_u} - W, \qquad (81)$$

$$\nu_u^{\mathrm{r}} \leq \overline{\nu}_u + \overline{W} \leq \overline{\nu}_u - \frac{1}{2L} + W \leq \tilde{\nu}_{l_u} + W, \qquad (82)$$

which implies

$$\mathcal{L}_u = \left[\nu_u^{\mathrm{l}}, \nu_u^{\mathrm{r}} \right] \subseteq \left[\tilde{\nu}_{l_u} - W, \tilde{\nu}_{l_u} + W \right] = \mathcal{M}(\tilde{\nu}_{l_u}). \qquad (83)$$

This completes the proof.

References

[1] M. Uysal and M. R. Heidarpour, "Cooperative communication techniques for future-generation HF radios," *IEEE Commun. Mag.*, vol. 50, no. 10, pp. 56–63, Oct. 2012.

[2] J. Wang, G. Ding, and H. Wang, "HF communications: Past, present, and future," *China Commun.*, vol. 15, no. 9, pp. 1–9, Sep. 2018.

[3] T. L. Marzetta, "Noncooperative cellular wireless with unlimited numbers of base station antennas," *IEEE Trans. Wireless Commun.*, vol. 9, no. 11, pp. 3590–3600, Nov. 2010.

[4] E. G. Larsson, O. Edfors, F. Tufvesson, and T. L. Marzetta, "Massive MIMO for next generation wireless systems," *IEEE Commun. Mag.*, vol. 52, no. 2, pp. 186–195, Feb. 2014.

[5] C. Sun, X. Gao, S. Jin, M. Matthaiou, Z. Ding, and C. Xiao, "Beam division multiple access transmission for massive MIMO communications," *IEEE Trans. Commun.*, vol. 63, no. 6, pp. 2170–2184, Jun. 2015.

[6] X. Yu, A.-A. Lu, X. Gao, G. Y. Li, G. Ding, and C.-X. Wang, "HF skywave massive MIMO communication," *IEEE Trans. Wireless Commun.*, vol. 21, no. 4, pp. 2769–2785, Apr. 2022.

[7] M. Chaudhary, N. K. Meena, and R. S. Kshetrimayum, "Local search based near optimal low complexity detection for large MIMO system," in *Proc. Int. Conf. Adv. Netw. Telecommun. Syst.*, 2016, pp. 1–5.

[8] S. Agarwal, A. K. Sah, and A. K. Chaturvedi, "Likelihood-based tree search for low complexity detection in large MIMO systems," *IEEE Wireless Commun. Lett.*, vol. 6, no. 4, pp. 450–453, Aug. 2017.

[9] S. Yang and L. Hanzo, "Fifty years of MIMO detection: The road to large-scale MIMOs," *IEEE Commun. Surveys Tut.*, vol. 17, no. 4, pp. 1941–1988, 4th Quart. 2015.

[10] M. A. Albreem, M. Juntti, and S. Shahabuddin, "Massive MIMO detection techniques: A survey," *IEEE Commun. Surveys Tut.*, vol. 21, no. 4, pp. 3109–3132, Fourthquarter 2019.

[11] A.-A. Lu, X. Gao, Y. R. Zheng, and C. Xiao, "Low complexity polynomial expansion detector with deterministic equivalents of the moments of channel gram matrix for massive MIMO uplink," *IEEE Trans. Commun.*, vol. 64, no. 2, pp. 586–600, Feb. 2016.

[12] I. A. Khoso, X. Zhang, I. A. Khoso, Z. A. Dayo, and X. Dai, "Computationally efficient data detection in massive MIMO wireless systems via semi-iterative method," *IEEE Trans. Veh. Technol.*, vol. 70, no. 10, pp. 10252–10264, Oct. 2021.

[13] L. Liu et al., "Energy- and area-efficient recursive-conjugate-gradient-based MMSE detector for massive MIMO systems," *IEEE Trans. Signal Process.*, vol. 68, pp. 573–588, 2020.

[14] S. Park, "Low-complexity LMMSE-based iterative soft interference cancellation for MIMO systems," *IEEE Trans. Signal Process.*, vol. 70, pp. 1890–1899, 2022.

[15] M. Mahdavi, O. Edfors, V. Öwall, and L. Liu, "Angular-domain massive MIMO detection: Algorithm, implementation, and design tradeoffs," *IEEE Trans. Circuits Syst. I.: Regular Papers*, vol. 67, no. 6, pp. 1948–1961, Jun. 2020.

[16] M. Abdelghany, U. Madhow, and A. Tölli, "Beamspace local LMMSE: An efficient digital backend for mmWave massive MIMO," in *Proc. IEEE 20th Int. Workshop Signal Process. Adv. Wireless Commun.*, 2019, pp. 1–5.

[17] T. Takahashi, A. Tölli, S. Ibi, and S. Sampei, "Low-complexity large MIMO detection via layered belief propagation in beam domain," *IEEE Trans. Wireless Commun.*, vol. 21, no. 1, pp. 234–249, Jan. 2022.

[18] D. Slepian, "Prolate spheroidal wave functions, Fourier analysis, and uncertainty–V: The discrete case," *Bell Syst. Tech. J.*, vol. 57, no. 5, pp. 1371–1430, May/Jun. 1978.

[19] H. A. A. Saleh, A. F. Molisch, T. Zemen, S. D. Blostein, and N. B. Mehta, "Receive antenna selection for time-varying channels using discrete prolate spheroidal sequences," *IEEE Trans. Wireless Commun.*, vol. 11, no. 7, pp. 2616–2627, Jul. 2012.

[20] M. Ruan, L. W. Hanlen, and M. C. Reed, "Spatially interpolated beamforming using discrete prolate spheroidal sequences," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process.*, 2006, pp. III-840–III-843.

[21] T. Zemen and C. F. Mecklenbrauker, "Time-variant channel equalization via discrete prolate spheroidal sequences," in *Proc. IEEE 37th Asilomar Conf. Signals, Syst. Comput.*, 2003, pp. 1288–1292.

Authorized licensed use limited to: Southeast University. Downloaded on January 18,2024 at 06:05:40 UTC from IEEE Xplore. Restrictions apply.

[22] O. Longoria-Gandara and R. Parra-Michel, "Estimation of correlated MIMO channels using partial channel state information and DPSS," *IEEE Trans. Wireless Commun.*, vol. 10, no. 11, pp. 3711–3719, Nov. 2011.
[23] E. Sejdic, M. Luccini, S. Primak, K. Baddour, and T. Willink, "Channel estimation using DPSS based frames," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process.*, 2008, pp. 2849–2852.
[24] J. K. Tugnait and S. He, "Multiuser/MIMO doubly selective fading channel estimation using superimposed training and Slepian sequences," *IEEE Trans. Veh. Technol.*, vol. 59, no. 3, pp. 1341–1354, Mar. 2010.
[25] T. Zemen and C. F. Mecklenbrauker, "Time-variant channel estimation using discrete prolate spheroidal sequences," *IEEE Trans. Signal Process.*, vol. 53, no. 9, pp. 3597–3607, Sep. 2005.
[26] G. Taubock, F. Hlawatsch, D. Eiwen, and H. Rauhut, "Compressive estimation of doubly selective channels in multicarrier systems: Leakage effects and sparsity-enhancing processing," *IEEE J. Sel. Topics Signal Process.*, vol. 4, no. 2, pp. 255–271, Apr. 2010.
[27] S. Sreepada and S. Kalyani, "Channel estimation in OFDM systems with virtual subcarriers using DPSS," *IEEE Commun. Lett.*, vol. 20, no. 12, pp. 2462–2465, Dec. 2016.
[28] L. E. Sweeney Jr, "Spatial properties of ionospheric radio propagation as determined with half-degree azimuthal resolution," Stanford, CA, USA: Stanford Electronics Laboratories, Tech. Rep., 1970.
[29] G. J. Frazer, "Experimental results for MIMO methods applied in over-the-horizon radar," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 32, no. 12, pp. 52–69, Dec. 2017.
[30] E. E. Johnson, E. Koski, and W. N. Furman, *Third-Generation and Wideband HF Radio Communications*. Norwood, MA, USA: Artech House, 2013.
[31] Z. Qin et al., "Link quality analysis based channel selection in high-frequency asynchronous automatic link establishment: A matrix completion approach," *IEEE Syst. J.*, vol. 12, no. 2, pp. 1957–1968, Jun. 2018.
[32] B. Wang, F. Gao, S. Jin, H. Lin, and G. Y. Li, "Spatial-wideband effect in massive MIMO systems," in *Proc. IEEE 23rd Asia-Pacific Conf. Commun.*, 2017, pp. 1–6.
[33] D. Shi, L. Song, W. Zhou, X. Gao, C.-X. Wang, and G. Y. Li, "Channel acquisition for HF skywave massive MIMO-OFDM communications," *IEEE Trans. Wireless Commun.*, vol. 22, no. 6, pp. 4074–4089, Jun. 2023.
[34] X. Gao, L. Dai, S. Han, C.-L. I, and X. Wang, "Reliable beamspace channel estimation for millimeter-wave massive MIMO systems with lens antenna array," *IEEE Trans. Wireless Commun.*, vol. 16, no. 9, pp. 6010–6021, Sep. 2017.
[35] X. Cheng, J. Deng, and S. Li, "Wideband channel estimation for millimeter wave beamspace MIMO," *IEEE Trans. Veh. Technol.*, vol. 70, no. 7, pp. 7221–7225, Jul. 2021.
[36] A. L. Moustakas, G. C. Alexandropoulos, A. Polydoros, I. Kaddas, and I. Dagres, "Impact of imperfect channel estimation in HF OFDM-MIMO communications," in *Proc. IEEE 30th Annu. Int. Symp. Pers. Indoor Mobile Radio Commun.*, 2019, pp. 1–6.
[37] D. B. Percival and A. T. Walden, *Spectral Analysis for Physical Applications*. Cambridge, U.K.: Cambridge Univ. Press, 1993.
[38] G. A. F. Seber, *A Matrix Handbook for Statisticians*, vol. 15. Hoboken, NJ, USA: Wiley, 2008.
[39] S. Karnik, Z. Zhu, M. B. Wakin, J. Romberg, and M. A. Davenport, "The fast Slepian transform," *Appl. Comput. Harmon. Anal.*, vol. 46, no. 3, pp. 624–652, 2019.
[40] E. Sejdic, M. Luccini, S. Primak, K. Baddour, and T. Willink, "Channel estimation using DPSS based frames," in *Proc. IEEE Int. Conf. Acoust., Speech, Signal Process.*, 2008, pp. 2849–2852.
[41] R. Hunger, "Floating point operations in matrix-vector calculus," Associate Inst. Signal Process., Tech. Univ. Munich, Munich, Germany, Tech. Rep., 2007.
[42] "Proplab-pro version 3, solar terrestrial dispatch," 2023. [Online]. Available: https://shop.spacew.com/index.php/product/proplab-pro-hf-radio-propagation-laboratory/
[43] G. Lane, *Signal-to-Noise Predictions Using VOACAP: A User's Guide*. Cedar Rapids, IA, USA: Rockwell-Collins, 2001.
[44] J. M. Steele, *The Cauchy-Schwarz Master Class: An Introduction to the Art of Mathematical Inequalities*. Cambridge, U.K: Cambridge University Press, 2004.
[45] M. A. Davenport and M. B. Wakin, "Compressive sensing of analog signals using discrete prolate spheroidal sequences," *Appl. Comput. Harmon. Anal.*, vol. 33, no. 3, pp. 438–472, Feb. 2012.



Linfeng Song (Graduate Student Member, IEEE) received the B.E. degree in radio wave propagation and antenna from Wuhan University, Wuhan, China, in 2020. He is currently working toward the M.E. degree with the National Mobile Communications Research Laboratory, Southeast University, Nanjing, China. His research interests include massive MIMO communications and HF communications.



Ding Shi (Graduate Student Member, IEEE) received the B.E. degree in information science and engineering from Southeast University, Nanjing, China, in 2018. He is currently working toward the Ph.D. degree with the National Mobile Communications Research Laboratory, Southeast University, Nanjing, China. Since November 2022, he has been a Visiting Student with the Division of Information Science and Engineering, KTH Royal Institute of Technology, Stockholm, Sweden. His research interests include massive MIMO communications, HF communications, and signal processing for wireless communications.



Lu Gan (Senior Member, IEEE) received the B.Eng. and M. Eng. degrees from Southeast University, Nanjing, China, in 1998 and 2000 respectively, and the Ph.D degree from Nanyang Technological University, Singapore in 2004. She is currently a Senior Lecturer with Brunel University London, Uxbridge, U.K. Before she joined Brunel University in 2008, she was on the faculties with The University of Newcastle, Newcastle, NSW, Australia, from 2004 to 2006 and University of Liverpool, Liverpool, U.K. from 2006 to 2007. From 2003 to 2004, she was a Research Associate with the Centre for Signal Processing, Nanyang Technological University, Singapore. Dr. Gan's research interests include fundamental signal processing theories and their applications in image/video coding and processing, non-destructive terahertz and ultrasound imaging, and machine learning and wireless communications. She is a regular Reviewer for many IEEE journals.



Xiqi Gao (Fellow, IEEE) received the Ph.D. degree in electrical engineering from Southeast University, Nanjing, China, in 1997. In April 1992, he joined the Department of Radio Engineering, Southeast University. Since May 2001, he has been the Professor of information systems and communications. From September 1999 to August 2000, he was a Visiting Scholar with the Massachusetts Institute of Technology, Cambridge, MA, USA and Boston University, Boston, MA. From August 2007 to July 2008, he was a Humboldt Scholar with the Darmstadt University of Technology, Darmstadt, Germany. His current research interests include broadband multicarrier communications, massive MIMO wireless communications, satellite communications, optical wireless communications, and information theory and signal processing for wireless communications. From 2007 to 2012, he was the Editor of the IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS. From 2009 to 2013, he was an Associate Editor for the IEEE Transactions on Signal Processing. From 2015 to 2017, he was the Editor of the IEEE TRANSACTIONS ON COMMUNICATIONS. Dr. Gao was the recipient of the Science and Technology Awards of the State Education Ministry of China in 1998, 2006 and 2009, the National Technological Invention Award of China in 2011, the Science and Technology Award of Jiangsu Province of China in 2014, and the 2011 IEEE Communications Society Stephen O. Rice Prize Paper Award in the field of communications theory.

Authorized licensed use limited to: Southeast University. Downloaded on January 18,2024 at 06:05:40 UTC from IEEE Xplore. Restrictions apply.